

Power Without Information: The Human Cost of Uniform Land Taxation in Subsistence Agriculture

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Abstract

A state that cannot verify the harvest it taxes chooses between a crop share, which invites a reporting game between the farm and the assessor, and a fixed rent per hectare, which cannot be moved by a report of the harvest and cannot follow it: a fixed rent taxes uniformly farms whose harvests differ. In January 1933 the Soviet state replaced its claim on the assessed grain harvest of collective farms with a fixed rent per sown hectare, one rate for every farm of a district. On the accounts of 272 collective farms of the Urals for 1933 and 1934, I measure what uniform taxation cost in people and divide the cost between information and incentives. The sequence of the agricultural year fixes what each party observed when it acted and hence the form of each claim. The rules estimated from the accounts, the district office's schedule, the farm's division of grain between seed and food, the seed relief and the response of population to food, convert any schedule of deliveries into members a year later. Relative to the same statutory total allocated on harvests with subsistence first, the policy as run cost 4,392 people, one member in twenty-eight: 707 from taxing uniformly farms whose harvests differed, 2,755 from the level of the wedge, the offices' collection above the norm, and 930 from its allocation across farms the office could not tell apart. Relief that came as seed in the spring rather than food in the autumn cost a further 964.

1 Introduction

A state that taxes an output it cannot verify at acceptable cost must choose the base of its claim. A crop share, a claim on the reported or assessed output, invites a reporting game: the producer understates the output and the assessor overstates it, and what the state collects is the joint result. A fixed rent on a base the state can verify, such as the area sown, removes the game, and it cannot follow the output: every unit whose base is the same owes the same, whatever its outcome. Mechanism design has priced the information a principal lacks in the transfers it pays for truthful reports (Hurwicz, 1972; Myerson, 1979; Baron and Myerson, 1982; Laffont and Tirole, 1993), and where transfers or commitment are absent it has derived rules with bounded discretion and escape clauses opened by verified hardship (Amador et al., 2006; Alonso and Matouschek, 2008; Halac and Yared, 2020). A coercive state pays no such transfers, commits only by statutes it must itself enforce, and holds the power to punish without limit, which relaxes the participation constraint and leaves the information constraint standing (Acemoglu and Wolitzky, 2011; Landa and Tyson, 2017); real authority over the amount taken then passes to whoever can see the output (Aghion and Tirole, 1997). What uniform taxation on a verifiable base costs such a state, and who bears the cost, has not been measured, because the two forms of claim are rarely observed on the same producer in the same year, and because the literature on the choice of contract form is voluntary, so that a fixed rent can always be attributed to the cost of enforcing a share rather than to the impossibility of seeing the output (Allen and Lueck, 2002; Coşgel and Miceli, 2005; Best et al., 2015).

Soviet grain procurement in 1933 and 1934 is the case in which both forms can be observed on one farm in one year and the cost of the fixed rent can be counted in people. Until 1932 grain was procured under a contract that was nominally a crop share. In practice each farm's quantity was apportioned down the hierarchy from a central plan on the basis of the assessed harvest and raised by a supplementary plan where the crop looked good. Both sides behaved strategically: the farm hid grain and understated its harvest, and the collector overstated it to fill his plan. The state could neither verify a report nor commit to it, since a harvest revealed in one year raised the next year's plan. The campaign of 1932 ran on this system and was followed by the famine of 1932–33 (Davies and Wheatcroft, 2004). On 19 January 1933 the statute moved the base of the claim from the harvest to the hectare. A collective farm now owed a fixed rent in grain per hectare sown, the norm, in the Urals about 1.8 centners, set before the sowing, one rate for every farm of a district, declared not subject to change; a supplementary plan became a criminal offence. Neither the farm's report nor the collector's assessment can affect such a claim, so the reporting game had no object.

The annual accounts of 272 collective farms of Sverdlovsk oblast for the harvests of 1933 and 1934 record, for the same farm and year, the harvest in the barn, the delivery the statute implied

and the delivery made, the fee paid to the machine-tractor station, the seed held and sown, the seed lent by the state, the grain distributed to the members, and the members a year later. Within a district the delivery did not vary with the farm's own harvest in either year. The two obligations that continued to be computed from the harvest, the station's fee and the flax contract, show the signs the reporting game predicts: the fee rose with the farm's harvest and was computed on an assessment above the barn, and the flax yield recorded in 1933 was a fifth of that recorded in 1934, with 34 farms that sowed flax recording no harvest at all. The delivery is consistent with less misreporting than the obligations that stayed on the harvest; the accounts do not show that misreporting was absent, and a fixed rent on land can still be reduced by understating the area, which the state inspected.

What the statute created in place of the reporting game was uniform taxation of farms whose harvests differed. At the 1933 median yield of 5.6 centners per hectare the norm took about a third of the harvest; on the farms with the poorest yields it took three fifths, and in 1933 the statute alone left a third of the farm-years below subsistence, 2,100 kilocalories per member per day. The district procurement offices, answerable for a district total under criminal liability, collected above the norm, between a fifth and a third above the statute at the median district in the two years; this collection above the norm is the wedge. Within a district the wedge varied with one thing only, whether the yield commission had certified the farm as failed: the statute on those farms, one multiple on all the rest. The state learned which farms the collection had left short only in the late winter, when it inspected the seed in every granary against the registered area, and in the spring it lent the missing seed back, 13,546 centners to 93 of the farm-years in the sample, 29 per cent of the wedge of 1933, at a charge of ten poods on every hundred. The loan restored the next sowing after the members had passed the winter on what the collection left.

These facts make the cost of the policy measurable, through two properties of the setting. The agricultural year ran in a fixed sequence, and the state's observations came at fixed points in it: the register in the spring, the assessment in the summer, the district's harvest during the collection, the seed stock in the late winter. The information available to each party at the time it acted therefore follows from the sequence, and it fixes the form of every claim: one rate per hectare from the centre, two values within a district from the office, a loan on the farm's own shortfall from the centre. With the reporting game removed and the information of each party given by the sequence, the model has two decisions, the office's collection above the norm and the centre's seed loan; the office's schedule can be rationalised as a corner solution and the loan as the restocking rule of a centre for which a restored hectare attracts the next year's rent. The second property is that the rules through which a delivery became food and food became people can be estimated from the accounts: the farm's division of its residual grain between seed and food, the relief that filled the shortfall of every farm that was short, and the response of the members a year later to food per

member, an elasticity of 0.135 on 252 farms, five eighths of it departure. These rules convert any schedule of deliveries into members a year later. Each counterfactual changes the allocation or the timing of the deliveries and holds the total collected in a district at the statutory total or at the total the office collected, so that what is measured is the cost of allocating a fixed rent on land rather than on harvests, and of repairing it late, not the cost of the norm relative to a different level of extraction.

Relative to the statutory total allocated on the farms' harvests with every farm held at subsistence where the total allowed, the policy as run cost 4,392 people over the two harvests, one member in twenty-eight. Taxing uniformly farms whose harvests differed cost 707 people, the information the centre gave up when it wrote the claim on the base it could verify. The wedge cost 3,685: 930 of them its allocation across farms the office could not tell apart, the office's information constraint, and 2,755 its level, the incentive constraint at the district. Relief that came as seed in the spring rather than food in the autumn cost a further 964, the information the centre obtained only from the seed stock. The cost concentrates where the claim crossed the subsistence line: the same norm cost half as much for the 1934 harvest, three fifths larger than that of 1933.

The contribution is threefold. The change of a tax base in a coercive state is observed at the level of the producer, with both forms of claim on the same farm in the same year, so that the removal of the reporting game is a prediction tested rather than an assumption. The cost of the information the state gave up is counted in people rather than in grain, with the relief the state paid to repair it, and divided among the information constraint of the centre, the information constraint of the office and the incentive constraint at the district. And the seed relief is shown to be a decision of the centre about the following harvest, sized on the one post-harvest quantity it could verify. The literature on the Soviet famine has studied the level of procurement across regions and its causes (Davies and Wheatcroft, 2004; Ellman, 2007; Naumenko, 2021; Markevich et al., 2025); the form of the claim at the farm, its allocation across farms the collectors could not tell apart, and its repair have not been observed before.

Section 2 describes the institutions and the statute of 1933. Section 3 describes the accounts. Section 4 reports the reduced-form evidence on the deliveries, the seed relief and the response of population to food. Section 5 sets out the model, Section 6 its estimation and fit, Section 7 the counterfactuals and the decomposition of the cost, and Section 8 concludes.

2 Institutional Background

The Soviet state of 1933 took grain from collective farms by administrative order. It set the quantity, fixed the price, and enforced payment by criminal law; the farms it took from were poor, and their members lived on what remained after the state's claims. Two claims of different form stood

on the grain of one farm in the same year: an obligatory delivery to the state, written from January 1933 on the hectares the farm had sown, and a fee to the machine-tractor station, written from February 1933 on a harvest assessed by a district commission. A third claim, on flax, stayed under contract. Figure 1 gives this structure in simplified form, the actors between the state and the farm and the claim each held with its base, the quantity from which the claim was computed; Figure 2 gives the sequence of one harvest year. The paragraphs below take the system in that order: the procurement it replaced, the obligatory delivery of 1933 and its stated purpose, the assessment of the harvest, the station's fee, the flax contract, the collective farm's funds and accounts, the harvest year, and the base of each claim.

On the 272 farms of Sverdlovsk oblast in the panel a farm held a median 394 hectares of grain and 251 members; grain yields were 5.6 centners per hectare in 1933 and 7.9 in 1934; after the 1933 harvest the median member received 1,718 kilocalories a day from the farm's distribution, and two farms in three fell below 2,100 (Section 3). Deliveries were paid at prices the state set (law of 19 January 1933, art. 1). Grain trade by collective farms was suspended from the first days of the deliveries (order of 20 June 1933, art. 1). The theft of collective grain was a capital offence under the law of 7 August 1932, and the internal passport introduced in December 1932 closed the towns to villagers without a permit (Davies and Wheatcroft, 2004).

Procurement before 1933. Through 1932 a collective farm's grain obligation was written as a contract, under which the farm undertook to deliver an agreed quantity and the procurement agency to supply credit, seed and agronomic service. The contract was not the operative instrument. The quantity a farm delivered was set by the state's procurement plan, apportioned from the centre to regions, districts, village soviets and farms, and regional and district authorities could raise the task handed to them to insure themselves against its non-fulfilment (Il'inykh, 2015, pp. 364, 366; Kondrashin, 2015, pp. 372–373). The plan could be raised during the year by a supplementary plan, formally a voluntary pledge and in practice a central decision apportioned downward, and a farm that had met its plan on a good harvest received an additional task at once (Il'inykh, 2015, pp. 366–367; Il'inykh and Simonov, 2021, pp. 10–11). For the 1932 harvest the Politburo required a safety margin of 4 to 5 per cent to be added to every district and farm plan (Davies and Wheatcroft, 2004). By 1932 the contract obligation resembled a tax whose final size was fixed only by the procurement plan (Moshkov, 1966, p. 200, cited in TSD III, p. 894).

The obligatory delivery. The law of 19 January 1933 abolished the contract system for grain and replaced it with an obligation “having the force of a tax”, paid at state prices (art. 1): a fixed quantity per hectare of the planned spring sowing and of the actual winter sowing, with separate norms for farms served and not served by a machine-tractor station (art. 2). A decree of 27 January

set the quantities by crop; in Ural oblast a farm without a station owed 1.9 centners per hectare of wheat, 2.3 of rye, 1.6 of oats and 0.8 of legumes, a farm with a station 1.5, 1.8, 1.3 and 0.6, and individual households 5 to 10 per cent more than the collective farms of their district (STO decree no. 78 of 27 January 1933). The republican, krai and oblast executive committees set and published a norm for each district by 10 February in the southern regions and 15 February elsewhere; the district norm could lie above or below the regional average and, once published, was binding on every farm of the district and not subject to change (art. 4). Each collective farm was handed a written obligation, the district norm applied to its area, by 15 March 1933 (art. 5). Non-delivery was punished by a fine for a collective farm and by prosecution for an individual household (art. 16). Local authorities and procurement organs were forbidden to allow supplementary plans or to impose any obligation above the per-hectare norms (art. 17); the order of 20 June 1933 made the issuing of a supplementary plan a criminal offence (art. 5), and the order of 2 August 1933 repeated the ban (item 4). The published norms contained a reserve of about 4 per cent above the quantity the state planned to receive, not mentioned in the law, from which the obligations of districts with bad weather could be reduced (Davies and Wheatcroft, 2004, pp. 254, 260). For the 1934 harvest the norms were set again by 15 February, separately for wheat, rye and legumes by republic, krai and oblast, and the obligations handed to farms by 1 April; no other delivery obligation could be handed to a farm (decree no. 389 of 10 February 1934, arts. 3–5).

The stated purpose. The law's preamble names its aims as raising yields, expanding the sown area of grain, and "the timely determination of firm obligations" (preamble). The order of 20 June 1933 contrasts the new system with the old: deliveries would be made "not according to changing plans but according to firm and unchangeable norms established by law", so that a farm "can firmly calculate its income", and "no counter-plans for the delivery of grain are henceforth to be allowed under any conditions" (preamble).

The assessment of the harvest. The state did not count harvests; it estimated them. The state yield commissions were created on 17 December 1932, a central commission and inter-district commissions subordinate to it (TSD III, doc. 338 and p. 917). Their instruction of 22 May 1933 told them to measure the crop on the root before the harvest, laying a metre frame on the field at intervals and threshing the cut ears by hand, and stated its purpose as forestalling the concealment of the harvest from the state (Zelenin, 1970, pp. 465–466). A draft decree of 19 June 1933 proposed district scales converting a verbal grade into centners per hectare and controlled harvests on two to three thousand farms (TSD III, doc. 338). The commission's deputy chairman described its figure as "not the barn harvest but the optimal harvest", with losses at harvest assumed at 10 per cent (doc. 356), and later admitted that for 1933 and 1934 the commission had "only orientational data

for some crops and threshing data from a limited number of collective farms” (TSD IV, doc. 274). The Commissar of Agriculture reported in September 1933 that where the sample squares gave a lower yield than the earlier forecasts the commission had added to them, and put the Ural yield of 1933 at 6.8 centners per hectare against 8.8 in 1930 (TSD III, doc. 357). For 1934 the commission put the biological yield for the USSR at 8.5 centners per hectare; its deputy chairman’s figure for the actual harvest was 7.3, and the collective farms’ own reports gave a barn yield of 6.3 (TSD IV, pp. 17–18). The commissions also checked the sown area: in one district the chairman of the inter-district commission prosecuted the head of the district land department for adding six thousand hectares of winter sowing on the instruction of the district executive chairman, who then demanded to review the commission’s reports and seized copies of the secret ones (TSD III, doc. 338). The commissions’ figures were secret from the districts (docs. 338; TSD IV, doc. 55).

The machine-tractor station fee. A machine-tractor station was a state enterprise that held tractors and threshers and worked the fields of the collective farms of its district. Through 1932 a farm paid the station in money at fixed rates per unit of work. Decree no. 346 of 5 February 1933 abolished both the fixed rates and the payment in money and set the fee as a percentage of the farm’s harvest, paid in grain (art. 9): 20 per cent for the full set of operations (art. 10), or by operation, spring ploughing and the raising of fallow at 10 per cent, autumn ploughing 9, sowing 3, mowing 4 and threshing 8 (art. 11). For each served farm the gross harvest was fixed, on the basis of the yield the inter-district commission had established for the district, by a district commission of the chairman or deputy chairman of the district executive committee, the district procurement plenipotentiary, the station’s director and the farm’s chairman (art. 12). The fee for work done before threshing was settled from the first batches of threshed grain and no later than one month after threshing began, the fee for threshing as threshing proceeded (art. 12); on the panel’s farms threshing and combining were about a third of the statutory fee, so most of the fee was charged before the farm’s grain had been counted. The head of grain procurement had asked four days before the decree for a fee under which a farm served by a station surrendered at least two poods per hectare more than an unserved one, enforced by a compulsory threshing charge (Chernov, memorandum of 1 February 1933). In June 1934 Kosior told the Central Committee plenum that the inter-district commissions’ data were secret and had no influence on the yield used for the fee, that “for the time being each station determines the yield in its own way”, and that the district commissions which were to assign the yield group did not yet exist (TSD IV, doc. 55). Decree no. 390 of 17 February 1934 replaced the percentage for field work by a quantity of grain per hectare of work graduated by the yield band the district commission assigned, and left threshing at 8 per cent and the combine at 11 per cent of the grain.

Flax. Flax was procured by a separate state agency, Zagotlen, under contract. The decree of 11 August 1933 states that flax and hemp, “unlike grain, meat, milk and potatoes”, are procured on the basis of contracts, enforceable in court against both sides (arts. 1–2); records that in the previous campaign deliverers had deceived the state with under-scuthed flax, and makes an obligation count as fulfilled only at contract quality, with a grain premium for higher quality (art. 4); and places the flax plan on the same district secretaries and station political officers who answered for grain (art. 8).

The collective farm. A collective farm, a kolkhoz, worked land and equipment held in common and paid its members by labour-days out of what remained after its obligations. The order of 2 August 1933 fixed the sequence: the farm paid the delivery, the station’s fee and the repayment of state loans, then filled its seed funds for the winter and spring sowing, an insurance seed fund of 10 to 15 per cent of its annual seed requirement and a fodder fund, and distributed the rest to members; no other fund was permitted (items 1–3). Its harvest appears in its own annual account, drawn up by its bookkeeper after the close of the economic year from the farm’s primary documents; the account carries the barn figure where the published statistics carried the commissions’ estimate. A third of collective farms filed such an account in 1930, a quarter in 1931 and two fifths in 1932 (Men’shikova, 2001). A statistical survey of May and June 1935, covering about one collective farm in twelve, checked the accounts of 1933 and 1934 against the farms’ primary documents (Men’shikova, 2001); the panel of 272 farms is described in Section 3.

The harvest year. Figure 2 follows one year. The norms were fixed in January and February and the obligations handed to farms by mid-March, before the spring sowing; the standing crop was assessed in July and August; deliveries began with the first threshing, the order of 20 June 1933 named July, August and September the decisive months, and grain trade by collective farms stopped in the Urals on 1 August (arts. 1–2); the station’s fee followed within a month of threshing; the seed and fodder funds were filled after the claims. Reductions of the obligation ran through a centrally rationed fund: on 15 September 1933 the Politburo set funds of reductions for named regions “out of the reserve of obligations handed to collective farms and individual households above the approved plan”, a reduction being carried into arrears collectible in the autumn of 1934 (Politburo protocol no. 145, items 63/46 and 92/76); between August and December such reductions were authorised on more than twenty occasions, and in September the Committee on Procurements reported that the Urals, with the two Volga regions and Kazakhstan, had exhausted the reserves in their plans (Davies and Wheatcroft, 2004, p. 262). Before the 1934 sowing the state checked the seed: the decree of 25 February 1934 ordered a germination test of all seed grain in every collective-farm granary, with samples from each lot, to be completed within ten days by the

district agronomists, the seed inspection and the members of the inter-district yield commissions, and released state seed to the farms both in exchange and as a seed loan (decree no. 391, arts. 2–3, 7, 9). The terms of a state seed loan are recorded for the North Caucasus and Ukraine in the decision of 18 February 1933: interest-free, with ten poods charged on every hundred at repayment for the state’s administrative and transport costs (TSD III, doc. 307, item 5); the panel records the repayment of the 1934 loans by Ural farms but not their terms.

The base of each claim. Table 1 records the base of each claim and the form of payment in each year. The delivery moved from a share of output to a quantity per sown hectare in January 1933 and stayed there; the station’s fee moved from a price per unit of work, paid in money, to a share of an assessed harvest in February 1933, and to a quantity per hectare of work by yield band in February 1934; flax stayed a contracted share of output throughout.

Table 1: The base of each claim, 1932–1934.

	Obligatory delivery: base	paid in	MTS fee: base	paid in	Flax: base	paid in
through 1932	output, share	grain	work performed, price	money	contracted output	fibre and seed
1933	land, centners per hectare	grain	assessed output, share	grain	contracted output	fibre and seed
1934	land, centners per hectare	grain	work, kg per hectare by yield band	grain	contracted output	fibre and seed

Notes. Delivery: SNK decree of 13 April 1930 through 1932; law of 19 January 1933 and STO decree no. 78 of 27 January 1933; decree no. 389 of 10 February 1934. MTS fee: fixed rates per unit of work in money through 1932; decree no. 346 of 5 February 1933, arts. 9–12; decree no. 390 of 17 February 1934, under which threshing (8 per cent) and the combine (11 per cent) stay proportional. Flax: decree no. 370 of 11 August 1933.

3 Data

The data are the annual accounts of 272 collective farms of Sverdlovsk oblast, in its boundary of January 1934, for the harvests of 1933 and 1934, 544 farm-years, taken from the primary forms of the statistical survey of May and June 1935 described in Section 2 and held at the State Archive of Sverdlovsk oblast (GASO, f. 1812, op. 2, dd. 154–159). Section 3.1 describes the form and the variables taken from it, Section 3.2 the checks applied to the accounts and the samples available for each part of the analysis, and Section 3.3 the constructed variables and the summary statistics.

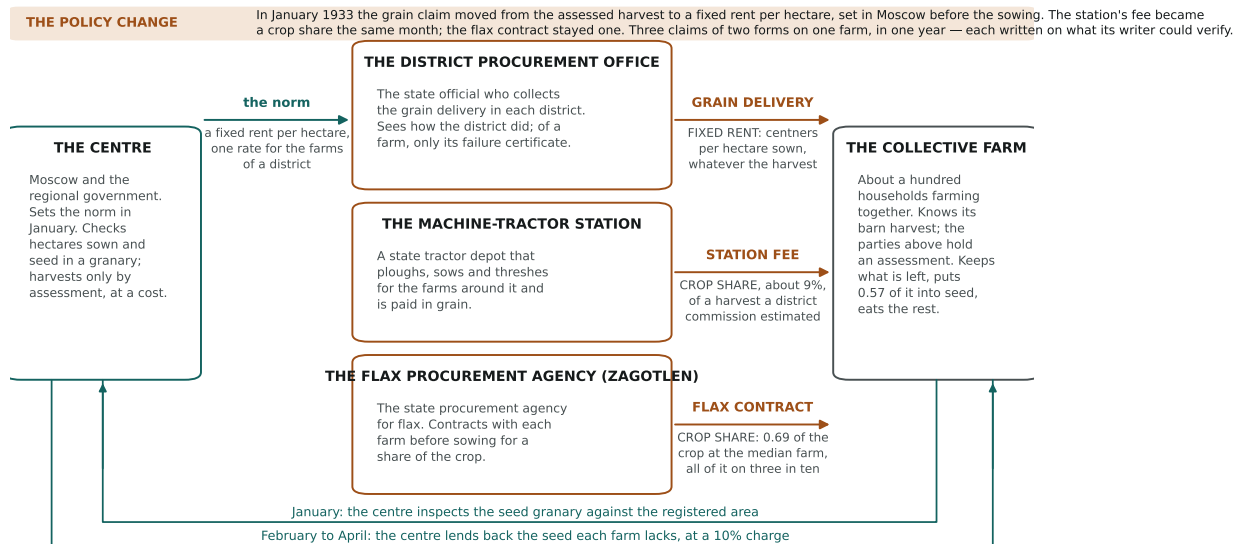


Figure 1: The actors between the state and the collective farm, and the claim each held with its base.

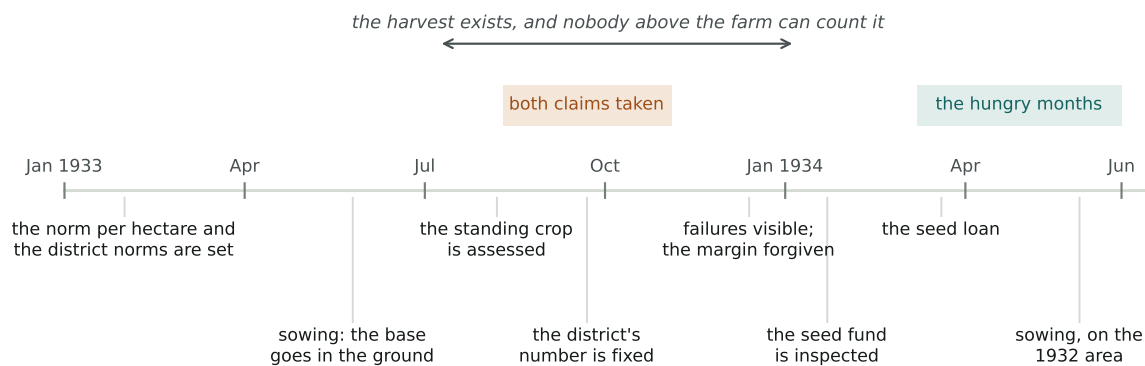


Figure 2: The sequence of one harvest year, 1933–34.

3.1 The accounts

Each form records one farm's economic position for one year on a single balanced sheet in eleven parts. The parts used here record the members and households on 1 January of the following year, with the households that joined and left during the year; the sown area, gross harvest and disposition of the harvest by crop; the work performed by the machine-tractor station, operation by operation, and the fee paid for it; the marketed output, including the obligatory delivery to the state; the distribution of the gross grain harvest among the state, the station, the farm's funds and its members; and the seed account of the following spring, with seed sown, seed held, seed lent by the state and seed purchased, by crop. The harvest is the farm's own figure, measured in the barn, and the disposition of the harvest is the farm's own record. Because both are entered on the same form, the delivery can be compared with the harvest and with the statutory obligation farm by farm.

3.2 The checks and the samples

The forms contain errors of transcription and arithmetic. Each farm-year was checked against the accounting identities of the form, that the gross harvest equals the sum of its uses and that the entries by crop sum to their totals; each discrepancy was compared with the scanned original and corrected where the original gave the figure. Entries that could not be corrected were marked as unusable. Thirty-one farm-years on 16 farms are unusable as a whole and are excluded from all of the analysis. Thirty-four farm-years have an unusable station sheet and are excluded from the analysis of the fee. Ten farm-years have an unusable delivery figure and are excluded from the analysis of the deliveries. Ninety-five farm-years have a ploughing figure that is derived rather than reported, which matters only where the station's work enters. The usable panel is 513 farm-years on 257 farms; the members of these farms numbered 74,011 on 1 January 1934 and 68,309 on 1 January 1935.

Table 2, panel A, gives the sample used in each part of the analysis. The deliveries are analysed on 493 farm-years of 248 farms in 50 districts; statistics computed within districts use the 70 district-years with at least three farms, in 36 districts, and the counterfactuals use the 440 of those farm-years with a complete record of food and members, holding 124,864 members over the two years. The seed relief is analysed on the 227 farms whose seed accounts reconcile. The population response is estimated on the 252 farms with members recorded at both dates and food in 1933. The station's fee is analysed on the farm-years that paid one, 245, of which 190 record the station's work per hectare and 70, in 1933, allow the assessed harvest to be recovered from entries that passed the checks. Flax is observed on the 231 farms that sowed it in either year.

3.3 Variables and summary statistics

The statutory obligation of a farm is the sum over crops of the published rate for the crop, at the rate for farms served or not served by a station, times the farm's sown area in that crop; the multiple of the statute is the delivery divided by this obligation. The yield is the gross grain harvest in the barn divided by the registered grain area. The seed requirement is the registered area times the seeding rate of 1.52 centners per hectare, and the seed shortfall is the excess of the requirement over the seed held. Food per member per day is the energy content of what the farm distributed to its members, the grain and legumes paid by labour-days, the potatoes and vegetables from the collective fields, and the milk of the household's own cow, at 3,400 kilocalories per kilogram of grain, 770 per kilogram of potatoes, 250 per kilogram of vegetables and 640 per litre of milk, the cow's yield being taken at 1,000 kilograms less the milk delivered to the state, divided by the number of members at 1 January and by 365. The measure excludes what a household may have bought or gathered, and it is an annual average, so that intake in the spring, before the new harvest, was below it. The subsistence line used throughout is 2,100 kilocalories per member per day. Each farm is assigned to its district on the administrative frame of 1928 and to the coordinates of its settlement; the distance from the farm to the district seat and to the nearest procurement point, and the settlements and population within ten kilometres, are used only as the farm characteristics of Section 4.1.

Table 2, panel B, reports the summary statistics by year. Between the two harvests the median farm's grain harvest rose by three fifths and its yield from 5.6 to 7.9 centners per hectare, while its delivery rose by six per cent; the delivery's share of the harvest fell from two fifths to a quarter. The seed fund was set aside at the same rate per hectare in both years. The grain distributed to members rose from 376 to 673 centners at the median farm, and food per member from 1,717 to 3,330 kilocalories a day; the share of farms below the subsistence line fell from two thirds to one in seven.

4 Reduced-Form Evidence

The farm accounts record, for each farm and each year, the grain delivered to the state and the sown area from which the statutory obligation can be computed, the seed held after the harvest and the seed lent by the state before the next sowing, the food distributed to members, and the number of members a year later. The regularities in these records that the model of Section 5 accounts for concern the deliveries, the seed loans and the members. Section 4.1 reports how much the district offices collected above the statute and how that collection varied within a district, across districts and between the two years. Section 4.2 reports which farms were short of seed, which received a

Table 2: Samples and summary statistics.

<i>A. Samples by use</i>		
Use	Sample	Where used
the deliveries	493 farm-years, 248 farms, 50 districts; 70 district-years with at least three farms in 36 districts	Sections 4.1, 6.2
the counterfactuals	440 farm-years in the 70 district-years, 124,864 members	Section 7
the seed relief	227 farms in 48 districts with reconciled seed accounts	Sections 4.2, 6.3
the population response	252 farms with members at both dates and food in 1933	Section 4.3
the station's fee	245 farm-years with a fee; 190 with the station's work per hectare; 70 in 1933 with a recoverable assessed harvest	Appendix B
flax	231 farms sowing flax in either year	Appendix B
<i>B. Summary statistics, usable farm-years, medians</i>		
Variable	1933	1934
gross grain harvest, centners	1,998	3,169
registered grain area, hectares	373	420
grain yield, centners per hectare	5.6	7.9
member households	63	60
obligatory delivery, centners	797	848
delivery as a share of the harvest	0.39	0.25
station's fee as a share of the harvest, farms that paid one	0.09	0.09
seed fund, centners per hectare sown	1.56	1.56
grain distributed to members, centners	376	673
distributed grain as a share of the harvest	0.17	0.22
food per member per day, kilocalories	1,717	3,330
share of farms below 2,100 kilocalories	0.66	0.15

Notes. Panel B: usable farm-years (513 on 257 farms), the number with each entry present ranging from 249 to 257; the harvest, area and yield on 250 and 252 farm-years with a positive harvest and area; food per member on 253 and 255 farm-years. The registered grain area pooled over both years has a median of 394 hectares, the figure used in Section 2. The harvest, delivery and distributed grain are grain and legumes. The fee share is computed on the farm-years that paid a fee, 99 in 1933 and 130 in 1934. The archival reference is to be confirmed before printing.

loan, and how the loan compared with the shortfall. Section 4.3 reports how a farm's membership a year later responded to the food its members received.

4.1 Collection above the Norm

The deliveries exceeded the statute almost everywhere. For each farm and year the accounts give the grain delivered under the obligatory delivery, and the statutory obligation, the statute for short, follows from the farm's sown area by crop and the published rates for farms served or not served by a machine-tractor station (Section 3). The ratio of the two, the multiple of the statute, is the measure used throughout, and the part of the delivery above the statute, the multiple less one, is the wedge in what follows. In 1933 the typical district collected about a third more than the statute from farms whose harvest had not failed, and in 1934 about a fifth more; in almost every district and year the multiple exceeded one, and the highest district collected one and a half times the statute (Table 3, panel A). The published rates are those for the Ural region as a whole. The norms that the oblast set for individual districts under the law are not among the sources available to me, so the multiple measures the collection above the regional rate and does not distinguish a district norm set above the regional rate from a collection above the district norm.

Within a district the deliveries took two levels. Figure 3 plots each farm's multiple against its own yield, by year, with the mean for each district in each year, hereafter a district-year, marked. Farms whose yield fell below about four centners per hectare delivered the statute and no more: their multiple is one at the median, whereas farms with yields just above four delivered a quarter above the statute. In the accounts a farm is treated as certified as failed when its barn yield is below four centners per hectare, except where the commission's own assessment can be recovered (Section 6.2). Below three centners per hectare the multiple falls below one, because the statute exceeded what the farm could deliver after seed and the remainder was carried into arrears (Section 2). Among all other farms the multiple does not vary with the farm's own harvest. Within a district-year the elasticity of the multiple to the farm's own yield is zero in both years (Table 3, panel B), and a farm that harvested twice its neighbour's crop delivered the same multiple of its statute. Farm characteristics that the office could observe from the district seat, the crop mix, the size of the farm, its distance from the seat and from the procurement point, and the population around it, explain six per cent of the variation in the multiple within districts; the only one with a measurable effect is the population within ten kilometres of the farm, and across its whole range that effect is a tenth of the difference in the multiple between districts.

Across districts the multiple rose with the harvest. Figure 4 plots the district-year multiple against the district's mean yield, and Table 3, panel C, reports the estimates. The elasticity of the multiple to the district's yield is about a third in 1933 and three tenths in 1934. Part of this is a

difference between districts that persists across the two years, which a norm set in February could reflect; part is a response to the year’s harvest, which it could not. The two parts can be separated because each district is observed in both years. The change in a district’s multiple between 1933 and 1934 rises with the change in its yield, by a quarter of the statute per log point, and falls with the change in the share of its farms that failed. In levels, too, the multiple falls with the share of failed farms. Had a district’s total been fixed independently of its failures, the multiple on its remaining farms would have risen with the failed share, since the same total would have been collected from a smaller base; the fall indicates that the district’s total was reduced where failure was certified. The number of collective farms in a district has no effect on the multiple.

The level at which failed farms delivered was itself one number per district, and it rose with the district’s multiple on the other farms, almost one for one, so that the ratio of the two levels was roughly constant across districts (Table 3, panel D). This is consistent with the failed farms having paid their district’s own norm, which under the law could be set above the regional rate in more productive districts and below it in less productive ones, with the office collecting a roughly constant margin above the district’s norm, about a tenth; it is equally consistent with the district’s norm equalling the regional rate and the failed level measuring the statute with error. The accounts do not separate the two, because the district norms are not observed; Section 6.5 returns to this.

Farms served by a machine-tractor station owed the lower statutory rate, and they delivered the same multiple of it as unserved farms in the same district and year (Table 3, panel E). Measured against the unserved rate instead, served farms appear to deliver a fifth less, which is the statutory discount and nothing else. The office collected the same proportion above the statute from both kinds of farm.

Because the obligation was fixed per hectare while harvests varied, the share of the harvest it took rose as the harvest fell. In 1933 the delivery took a quarter of the harvest on farms with yields above eight centners per hectare, a third between six and eight, close to half between three and five, and three fifths on farms below three, which paid less than the statute and still gave up most of what they grew (Figure 3, lower panel).

Table 3: The multiple of the statute, in five panels: the distribution of the multiple by year (A); the within-district elasticity to the farm’s own yield and the joint contribution of farm characteristics (B); the district-level regressions in levels, in persistent and transitory components and in changes between the two years, with the failed share and the number of farms (C); the failed level on the district’s multiple (D); and the comparison of served and unserved farms, at their own rates and at a common rate (E).

to be built from the run outputs

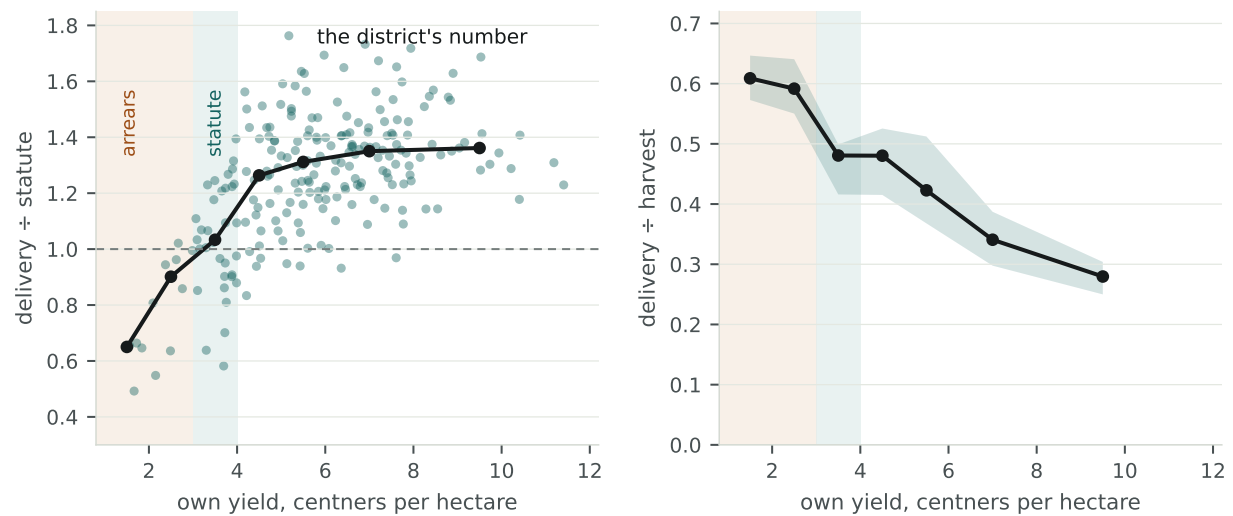


Figure 3: The farm's multiple of the statute against its own yield, with district-year means, 1933 and 1934, and the share of the harvest delivered by yield.

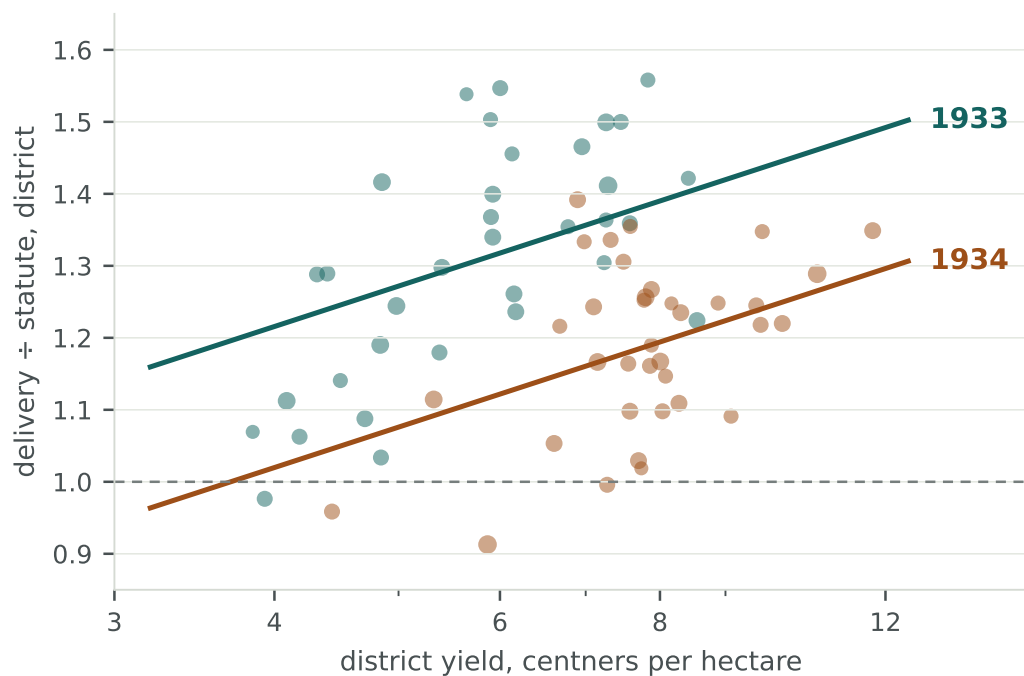


Figure 4: The district-year multiple against the district's mean yield, in levels and in changes between the two years.

4.2 Seed Relief

After the 1933 harvest many farms held too little seed to sow their registered area. A farm's seed requirement is its registered area, the area entered in the land register before the harvest and used as the plan for the next sowing, times the standard seeding rate, and its seed shortfall is the excess of that requirement over the seed fund it held; two farms in five were short. Before the spring sowing the state lent seed, and the loans are recorded in each farm's accounts. Figure 5 plots each farm's loan against its shortfall, and Table 4 reports the estimates discussed below.

The loans matched the shortfalls. Four fifths of the farms that were short received a loan, and at the median farm the loan equalled the shortfall; in total the loans covered the shortfalls. The farms left without a loan were those with the smallest shortfalls: the share relieved rises from two thirds in the lowest quartile of the shortfall to nine tenths in the third. The loans thus restored the farms that were short to their registered area, and only the farms whose shortfall was small were left out. The relief was seed rather than food, seed being five sixths of the grain lent, and it went to the farms whose harvest had failed: three quarters of the farms in the lowest third of the 1933 yield distribution received a seed loan, against one quarter of those in the highest third. In total the state lent back about 9 per cent of what these farms had delivered in 1933. Some farms received seed although their fund covered their registered area; the area sown in 1934 exceeded the registered area, and part of the relief financed sowing beyond it.

The loans were set farm by farm. Within a district a farm's loan rises almost one for one with its own shortfall (Table 4, column 2). Had each district received an allocation of seed to divide among its farms, a farm would have received less when its neighbours needed more; the shortfalls of the other farms in the district have no effect on a farm's loan (column 3). The loan was set on the farm's own seed account.

The state's other seed transaction of the same weeks shows that this was a rule on grain seed, not a general distribution. The state sold flax seed to the farms that were to sow flax. The flax seed a farm acquired followed its flax area and bore no relation to its grain seed shortfall, while the grain loan followed the grain shortfall and bore no relation to the flax area (columns 4 and 5).

The relief restored the sowing. Among the farms short of seed, those that received a loan expanded their grain area between 1933 and 1934 by 8 hectares on average, net of land released from flax, while those that received none reduced theirs by 46. Holding the shortfall fixed, a loan equal to the shortfall left the area unchanged, as it should if seed determines the area sown. Work by the machine-tractor stations does not account for the difference: farms short of seed were relieved at the same rate whether or not a station served them, and station ploughing explains four per cent of the difference in area between relieved and unrelieved farms.

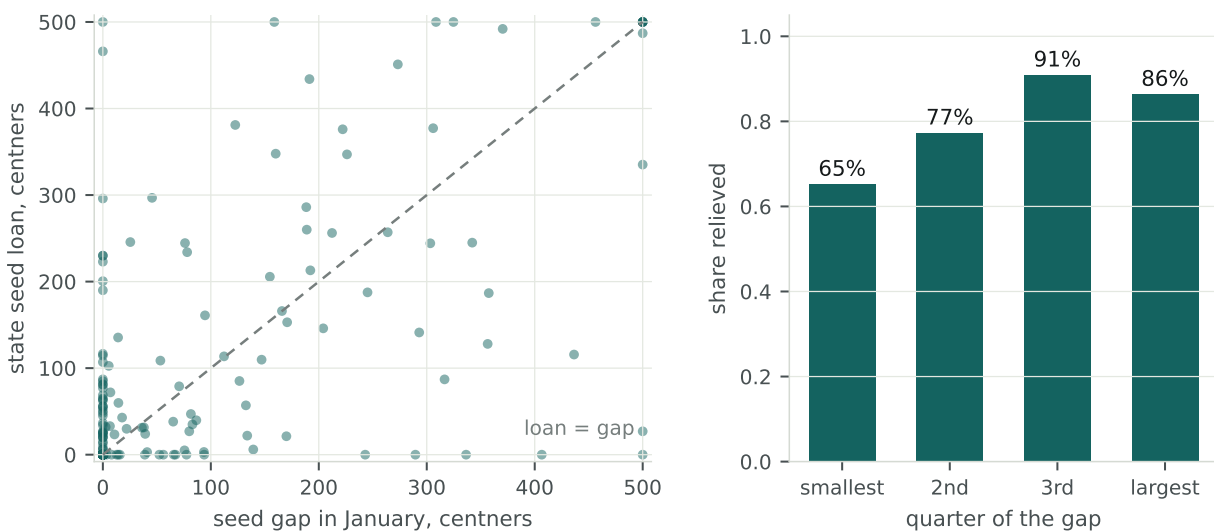


Figure 5: The loan against the shortfall farm by farm, with the share relieved by quartile of the shortfall.



Figure 6: Seed sown per hectare in 1934 by source and by tercile of the 1933 yield.

Table 4: The loan per hectare on the farm's own shortfall without and with district fixed effects, with the neighbours' shortfalls, and the grain loan and the flax seed each on the grain shortfall and the flax area.

to be built from the run outputs

4.3 Food and Population

A farm that kept less food had fewer members a year later. The accounts give the number of members of each farm on 1 January 1934 and on 1 January 1935, and the households that joined and left during 1934; the food the farm distributed to its members from the 1933 harvest, in grain, potatoes, vegetables and milk, is converted to kilocalories per member per day as described in Section 3. The calendar year 1934 was fed by the 1933 harvest, since the new crop was not in before its summer, so the change in members over 1934 is the population response to the food of 1933. Table 5 reports the estimates.

The response is estimated in levels, the number of members at the start of 1935 on the food per member of 1933, holding the number of members at the start of 1934 fixed, so that no denominator is shared between the two sides. A farm with ten per cent more food per member had 1.35 per cent more members a year later, and the response is the same when it is estimated within districts. On the narrower measure that counts grain alone the response is slightly larger. Yield and soil are not held fixed, because the food a farm kept was built from its harvest; conditioning on them would remove the variation that identifies the response.

The response is a loss of people, and most of it was departure rather than death. Netting out the households recorded as joining and leaving during 1934 leaves the natural change, births net of deaths. Of the population response to food, about three eighths is natural decrease and five eighths is departure. The proportion is consistent with the region's aggregate vital statistics for 1934, computed from a different source, which put rural natural change for the year close to zero while the panel's farms lost a twelfth of their members. The Urals had an industrial labour market at their door, and hungry households left for it despite the passport system introduced to stop them (Section 2). The measure is therefore a population response rather than a death rate, and every human cost in Section 7 is divided into deaths and departures in the same proportion.

The response does not reflect places that had always lost people. Food per member in 1933 is uncorrelated across districts with the rural death rate of 1927 to 1929, with its trend, and with infant mortality in those years, and the response is unchanged when those baseline rates are added as controls. It is correlated with the natural increase of the 1920s, districts that were growing faster then having kept more food in 1933; conditioning on that as well lowers the elasticity by about a tenth and leaves it well determined.

Table 5: The number of members at the start of 1935 on food per member in 1933, without and with district fixed effects, on the total and the grain-only measures; the decomposition of the raw response into natural change and departures; and the specifications with the 1927 to 1929 baseline rates added.

to be built from the run outputs

5 The Model

The model takes the norm as given and asks what the state's remaining decisions were under it. Because the delivery was computed from sown area, neither the farm nor the collector could change it by misreporting the harvest (Section 2; Appendix B), so the model contains no reporting game and treats the harvest recorded in the accounts as the harvest. What remains is the district office's decision, how much to collect above the norm and from which farms, and the centre's, how much seed to lend to a farm found short before the next sowing. The model writes down both problems, shows that the two rules documented in Section 4 can be rationalised as their solutions, and defines the outcomes through which any delivery rule, observed or counterfactual, is converted into a number of people in Section 7. Section 5.1 states the assumptions, 5.2 the primitives, and 5.3 the sequence of the year and the information of each agent; Sections 5.4 to 5.6 solve the problems of the office, the farm and the centre; Section 5.7 defines the outcomes.

5.1 Assumptions

Assumption 1 (No reporting game under the norm). *The obligatory delivery is computed from the registered sown area and the district norm. Neither the farm's report of its harvest nor an assessment of it enters the obligation. The farm therefore has no incentive to understate its harvest and the office none to overstate it, and the harvest recorded in the farm's barn is taken as its harvest.*

The assumption rests on the statute and on the form of the obligation. The law of 19 January 1933 based the delivery on the hectares of planned spring and actual winter sowing, and the yield commissions checked the sown area (Section 2). Appendix B shows that an obligation computed from an assessed harvest gives the farm a reason to understate the harvest and the assessor a reason to overstate it, and that an obligation computed from sown area gives neither any reason to misreport, whatever the rate and the penalties (Proposition B.1). The accounts are consistent with this: within a district the delivery did not vary with the farm's own harvest (Section 4.1). The two obligations that continued to be computed from the harvest, the station's fee and the flax contract, show signs of the misreporting the proposition predicts, but the samples are small and the assessed harvest has to be reconstructed, so the size of that misreporting is not measured (Appendix B). An obligation computed from land can be reduced by understating the sown area; the registered area is treated as correct because the state inspected it and based the following year's sowing plan on it. Farms served by a station owed a fee equal to a small share of their assessed harvest, which gave them some incentive to understate the harvest in the barn, so their recorded harvest may be slightly understated.

Assumption 2 (Independence of the two obligations on grain). *The delivery and the station's fee are computed from different bases by different bodies, and neither responds to the other at the level of the farm.*

The statutes fix the rate of each obligation and the order in which the two are paid, the delivery first and the fee second (order of 2 August 1933). In the accounts, a farm served by a station and a farm not served, in the same district and year, delivered the same multiple of their respective statutory obligations (Section 4.1), and on the farms whose fee was computed from an assessed harvest the multiple delivered bears no relation to the assessment (Appendix B). The delivery and the fee therefore enter the model separately, and the model contains no interaction between the office and the station.

5.2 Primitives

The model has three decision-makers: the centre, the state's central organs, which fixes the norm and lends seed; the district procurement office, which collects the deliveries of the farms in its district; and the collective farm, which divides what is left to it between seed and food. Two other bodies act during the year but make no decision in the model: the yield commission, which certifies the farms whose harvest failed and fixes the assessed harvest of the farms served by a machine-tractor station, and the station itself, whose fee is taken as recorded. The quantities below are taken from the statutes and the farm accounts or estimated from the panel; Table 7 in Section 6.1 gives their symbols, values and sources. The variables the agents choose are introduced with their problems in Sections 5.4 to 5.6.

The farm. Farm i in district D has a registered grain area $\bar{a}_i$, in hectares, sown in the spring and entered in the land register before the harvest. This area is the base of the delivery, and it is the state variable of the centre's problem. The farm's grain yield y_i , in centners per hectare, is a random variable. Its logarithm is the sum of a district-year component, a permanent farm component and a transitory component; the variances of the two farm components are recovered from the dispersion of log yield net of district-year effects and from its correlation across the two years. The harvest is $H_i = y_i \bar{a}_i$, measured in the farm's barn (Assumption 1). Sowing one hectare requires ς centners of seed. In the accounts, seed sown is proportional to area sown, so the seed a farm holds in the spring determines the area it can sow; this proportionality is the link between the grain a farm keeps in one year and its area in the next. When the grain left to a farm after its obligations, which I call the residual, is insufficient for both its seed and its food, the farm puts a share φ of it into seed; Section 5.5 states the rule. The share φ is estimated from the farms that were short of both. The order of 2 August 1933 required every farm to hold an insurance seed fund of 10 to 15 per cent of

its annual seed requirement in addition to its seed fund; κ denotes this margin. The farm has N_i members. Its distribution of grain, potatoes, vegetables and milk to them gives food per member per day, k_i , in kilocalories, computed as described in Section 3.

The obligations. The obligatory delivery is n_i centners per registered hectare. The norm n_i is the average of the statutory rates by crop over the farm's own sown area, at the rates for farms served or not served by a station (Section 2). The district office is handed a total T_D for its district. The model treats T_D as given; the accounts do not identify how its level was set (Section 6.5). Farms whose yield fell below a threshold y_F were certified as failed by the yield commission and were required to deliver the statutory amount only. In the accounts the threshold appears as a break in the multiple of the statute at about four centners per hectare (Section 4.1); the model takes it as given. A farm served by a station owes a fee equal to a share m of its assessed harvest, where m is the sum of the statutory percentages of decree no. 346 for the operations the station performed. The model does not explain the fee. It takes each farm's fee at its recorded value, and in the centre's problem an additional hectare attracts the fee at the average share observed on served farms.

The centre. Seed lent to a farm was repaid with a charge of ten poods on every hundred poods lent, the term recorded for the North Caucasus and Ukraine in the decision of 18 February 1933; the panel records the repayment of the Ural loans but not their terms (Section 2). The model takes the cost to the centre of one centner of seed restored to a farm as $1 + \lambda$ centners, with $\lambda = 0.10$ equal to this charge. A hectare restored to a farm enters its later obligations through the register, because the delivery was computed from the planned sowing and the plan followed the area last recorded (Section 2): the sowing plan for 1934 was based on the area recorded in 1932. The response β of the next year's base to a restored hectare is the elasticity of a district's sowing plan to its recorded area, estimated across districts. Part of the increase in grain area between 1933 and 1934 came from land released from flax; g_f denotes the hectares so released per hundred hectares of the grain plan, measured from the farms that reduced their flax area, and it enters the transition of the area from one year to the next. The centre discounts the following year's collection by a factor δ . This factor is not observed. Section 6.3 derives a lower bound on it from the observed seed relief.

Outcomes. The subsistence line is 2,100 kilocalories per member per day (Section 3). The number of members a year later responds to food per member with elasticity ε , estimated in Section 4.3 together with the shares of the response accounted for by deaths and by departures. A centner of grain not taken from a farm adds $\pi \times 3,400 \times 100$ kilocalories to the farm's food over the year, at the energy content of grain used in Section 3. In the main results $\pi = 1$, so that a centner not taken

is a centner distributed to members. Lower values allow for the part of such a centner that a farm would have put into its seed and fodder funds instead; Section 7.4 reports the results at $\pi = 0.6$ and at $\pi = 1 - \phi$, the share that the farm's own division rule leaves for food where its seed constraint binds.

5.3 Information and Timing

Information enters the model as a restriction on how finely each agent can differentiate among farms, not as a constraint on what farms report. By Assumption 1 no report of the harvest enters any obligation on grain, so there is no truth-telling constraint to impose. What remains is that an agent can set different amounts for two farms only if it can tell them apart at the time it acts. The obligations and the relief are therefore functions of what the agent that set them observed at the time, and the dates matter as much as the observations. The sequence of one harvest year and the information of each agent at the time it acts imply that every correction of an obligation came after the obligation had been collected, and that the centre's information about a farm was finer than the office's but arrived later and concerned seed alone.

The sequence of the year. Figure 2 in Section 2 gives the dates. In January and February the centre fixes the norm, and the obligations are handed to the farms by mid-March in 1933 and by 1 April in 1934; the district office receives the total for its district in the same period. In May the farm sows and its area is registered. In July and August the yield commission assesses the standing crop, and farms whose crop failed are certified as such. From the first threshing to the autumn the office collects the delivery and the station collects its fee; the fee for work done before threshing was settled from the first threshed grain, within a month of the start of threshing (Section 2). In the winter the farm fills its seed fund from what is left and distributes the remainder to its members. In the late winter the state inspects the seed in every granary against the registered area, and in the weeks before the spring sowing it lends seed to farms that are short. The next spring the farm sows what its seed allows. The members counted a year later reflect the food of the winter. In the notation of the model, the year runs from the norm n to the loan r , and the state of a farm at the start of the next year is the area it can sow.

What each agent observes. The centre, when it fixes the norm, observes the registered area of each farm and whether a machine-tractor station serves it, and nothing about a harvest that has not yet been sown. The norm can therefore differ across districts, as the law allowed, and across farms only in proportion to their area and according to whether a station serves them. When the centre lends seed, it observes for each farm the seed in its granary against its registered area, that is, the

seed shortfall σ_i . It does not observe the farm's harvest or the food its members received. The district office, when it collects, observes the harvest of its district as the collection proceeds, and for each farm whether the commission certified it as failed. It does not observe the harvest of an individual farm. The office's plenipotentiary sat on the commission that fixed the assessed harvest of each farm served by a station (Section 2), so for those farms an assessment existed; Section 4.1 shows no sign that the office used it, since served and unserved farms delivered the same multiple of their statutes. The model therefore gives the office two categories of farm within a district, failed and not failed, and lets its collection depend on the district's harvest. The district commission and the station observe, for each served farm, the assessed harvest $\hat{H}_i$, and the station observes in addition the grain it threshed itself, work that accounted for about a third of the statutory fee in 1933. The farm observes its own harvest. It sets no obligation and divides what is left to it by the rule described in Section 5.5.

The formal restriction. Let P_j denote the partition of the farms of a district into the groups that agent j can distinguish when it acts. Any quantity that agent j sets for individual farms must be constant within each group of P_j . For the office, P_{office} has two groups, the failed farms and the rest, so the delivery it collects can take at most two values within a district beyond the statutory proportionality to area; Section 5.4 shows that it takes exactly two. For the centre at the time of the loan, P_{centre} distinguishes farms by their seed shortfall, so the loan can be a function of σ_i and of nothing else about the farm. For the commission and the station, the assessed harvest is a separate number for each farm, so the fee varies farm by farm.

Corrections after collection. Both corrections of the grain obligations, the office's waiver of its excess on farms certified as failed and the centre's seed loan, occur after the obligation they correct has been collected, and each is based on an observation different from the one on which the obligation was set: the office's on the commission's certificate, the centre's on the seed stock. An observation made after an obligation has been collected can be used to return grain to a farm; it cannot be used to set the obligation. The statutory delivery of a farm whose harvest failed was collected in the autumn at the same rate per hectare as its neighbour's, and the seed that returned to it came before the next sowing. This is the timing component of the cost measured in Section 7.

The centre's information. The centre's information about a farm in the late winter is finer than the office's in the autumn. The office distinguished two kinds of farm; the seed shortfall distinguishes every farm that is short. The centre did not know less than the office. It learned later, and about one consequence of the harvest only, the seed the farm had left. The state inspected seed and not food (Section 2), and the relief accordingly took the form of seed.

An illustration. Consider three farms in one district, each with 100 registered hectares, with harvests of 300, 500 and 700 centners. Let the norm be 2 centners per hectare, so that each owes 200 centners by statute, and let the office collect one quarter above the statute from farms not certified as failed. The first farm is certified as failed and delivers 200; the second and the third each deliver 250, although their harvests differ by 200 centners. Their residuals are 100, 250 and 450 centners. Suppose each farm needs 150 centners of seed to sow its area, holds a seed fund of 165 centners when it can, and puts half of its residual into seed when the residual covers neither seed nor food. The first farm puts 50 centners into seed and 50 into food and is 100 centners short of seed; the second puts 125 into seed and 125 into food and is 25 short; the third fills its fund of 165 and consumes 285. In the autumn the office set two values, 200 and 250. In the late winter the centre observes three seed shortfalls, 100, 25 and 0, and lends 100 and 25. The food of the three farms over the winter, 50, 125 and 285 centners, is observed by nobody above the farm, and the loan does not change it.

Table 6: Information of each agent at the time it acts.

Agent	Acts	Observes at that time	Its decision can depend on
Centre, fixing the norm	January–March, before sowing	registered area; whether a station serves the farm; the district	area, station service and district: $n_i \bar{a}_i$
Yield commission and station (actions taken as given)	July–August (assessment); autumn (fee)	the standing crop of each served farm; grain threshed by the station	a separate assessed harvest $\hat{H}_i$ for each farm: the fee M_i
District office	autumn	the district’s harvest; for each farm, whether certified as failed	two values within a district and the district’s harvest
Farm	winter	its own harvest and residual	its division of the residual between seed and food
Centre, lending seed	late winter, before sowing	the seed in each granary against the registered area: σ_i	the seed shortfall of each farm: the loan r_i

Notes. Dates from the statutes and orders cited in Section 2. The share of the statutory fee accounted for by threshing and combining is computed from the recorded station work on the panel’s fee payers in 1933 (Section 3).

5.4 The District Office

Section 4.1 documented the regularities in the deliveries that a model of the office must account for. Within a district the multiple of the statute took two levels: the statute itself on farms whose harvest had failed, and one common multiple on all other farms, whatever their harvests. The common multiple rose with the district’s harvest, in levels and in the change between the two years, and fell with the share of the district’s farms that had failed. And served and unserved farms delivered the same multiple of their respective statutes. The problem written down here for the

district procurement office can account for such a schedule. Its form follows from the office's information, which restricts the schedule to two values within a district; from its liability for the district's total, which fixes the sum of the deliveries; and from a preference for lightening the burden on farms whose harvest failed, which holds for any positive weight the office placed on the district's future and sets the delivery of a failed farm at its statutory obligation. What remains is one free number per district: the multiple of the statutory obligation collected from farms not certified as failed. The model accounts for the form of the schedule and the treatment of failed farms; the level of the district's total is taken as given.

The objects. Let $A_D = \sum_{i \in D} n_i \bar{a}_i$ denote the statutory total of district D , the sum of its farms' obligations under the norm. Let F_D be the set of farms in the district that the yield commission certified as failed, those whose assessed yield fell below y_F , and V_D the remaining farms; A_F and A_V denote the statutory totals of the two groups. Before the harvest the office is handed a total T_D for its district, which in the accounts exceeded the statutory total in most districts in both years (Section 4.1). The office may not reduce a farm's delivery below its statutory obligation; reductions below the statute were granted by the centre through a separate fund (Section 2).

The schedule. By Section 5.3 the office distinguishes two groups of farms within its district. The most general schedule of deliveries it can set is therefore

$$x_i = \mu_D n_i \bar{a}_i \quad \text{for } i \in V_D, \quad x_i = \zeta_D n_i \bar{a}_i \quad \text{for } i \in F_D, \quad \mu_D \geq 1, \quad \zeta_D \geq 1, \quad (1)$$

where μ_D is the multiple of the statutory obligation collected from farms not certified as failed and ζ_D the multiple collected from failed farms. Within each group the delivery is proportional to the statutory obligation, so a served and an unserved farm in the same group pay the same multiple of their respective obligations (Assumption 2). The total collected is $X_D = \mu_D A_V + \zeta_D A_F$.

The office's problem. The office chooses the two multiples μ_D and ζ_D , and through them the deliveries in (1) and the total $X_D = \mu_D A_V + \zeta_D A_F$. Its payoff is the gain from the grain collected, less the penalties attached to the district's total, less the value it may place on the district's future sowing. It gains b per centner collected. It pays a penalty q per centner by which the total falls short of T_D , the criminal liability for an unfulfilled plan, and a penalty q_O per centner by which the total exceeds T_D , the prohibition of supplementary plans (Section 2); both penalties exceed b . And, to allow for the possibility that the office cared about the district's future deliveries, it may value the area each farm will be able to sow in the following year, a'_i , through the function $v(a)$ defined in Section 5.6, with a weight $\delta_O \geq 0$. By the division rule of Section 5.5, a centner taken from

a farm that is short of seed reduces the area it can sow next year by φ/ζ hectares, and a centner taken from a farm whose seed fund is full leaves its area unchanged. The office's problem is

$$\max_{\mu_D, \zeta_D} bX_D - q(T_D - X_D)^+ - q_O(X_D - T_D)^+ - \delta_O \sum_{i \in D} [v(\bar{a}_i) - v(a'_i)], \quad \text{subject to } \mu_D \geq 1 \text{ and } \zeta_D \geq 1. \quad (2)$$

The constraints are the statutory floor, since the office may not reduce any farm's delivery below its statutory obligation, and the restriction to two values already built into (1).

The solution. The problem separates into the choice of the total and the choice of its division, and both have corner solutions. Whatever the division, the payoff rises with X_D at the rate $b + q$ per centner while X_D is below T_D and falls at the rate $q_O - b$ once X_D is above it; the future term, which is non-increasing in every delivery, only strengthens the fall. The payoff therefore reaches its maximum at the kink, $X_D = T_D$. The office collects exactly its total, and the first-order condition for the total holds as an inequality on either side of the kink, not as an equality. With the total fixed at T_D , the choice of ζ_D determines $\mu_D = (T_D - \zeta_D A_F)/A_V$. Moving one statutory centner of collection from the failed farms to the others leaves the current payoff unchanged, because the total is unchanged, and changes the future term by the difference between the marginal value of a hectare on the failed farms and on the seed-short farms among the others. The derivative of the payoff with respect to ζ_D , with the total held fixed, is

$$-A_F \delta_O (\varphi/\zeta) [\bar{\mathbb{E}}_F(v'(a'_i)) - \bar{\mathbb{E}}_V(v'(a'_i) \chi_i)],$$

where $\bar{\mathbb{E}}_F$ and $\bar{\mathbb{E}}_V$ denote averages per statutory centner over the failed farms and over the other farms, and χ_i equals one on farms short of seed. Every failed farm is short of seed and will sow less next year than any other short farm, and v is concave, so the bracket is positive and the derivative is negative for every $\delta_O > 0$. The office therefore lowers ζ_D until the statutory floor stops it, $\zeta_D = 1$, and the multiple on the other farms follows from the total, $\mu_D = (T_D - A_F)/A_V$. At $\delta_O = 0$ exactly the derivative is zero and the office is indifferent among all divisions of its total.

Proposition 1 (The office's schedule). *Let v be concave, let q exceed b plus the largest marginal future value of a centner, and let every farm certified as failed be short of seed and able to sow less next year than any other farm that is short. Then: (i) the office collects exactly its total, $X_D = T_D$; (ii) for every $\delta_O > 0$ the office collects the statutory obligation and no more from farms certified as failed, $\zeta_D = 1$, and the multiple on the remaining farms is $\mu_D = (T_D - A_F)/A_V$; (iii) the condition that determines ζ_D does not depend on δ_O .*

The proof, which formalises the argument above with a Lagrange multiplier on the total, is in Appendix A.

What is left to estimate. Both decisions are corner solutions, so the parameters of the office's payoff, b , q , q_O and δ_O , do not appear in the schedule and cannot be recovered from it: the total is T_D whatever their values, and the division is the same for every positive δ_O . What the schedule leaves is two testable features, that farms certified as failed deliver the statute and that all other farms in a district deliver one common multiple, and one free number per district, T_D , which is taken from the accounts. Section 6.2 tests the two features and describes the level of T_D .

A one-period problem. The accounts show the pattern of a corner solution: farms certified as failed delivered their statutory obligation, and all other farms in a district delivered the same multiple of theirs (Section 4.1). Given the district's total, this schedule leaves the office no further choice. Proposition 1 shows that the corner is the solution for any positive weight the office might have placed on the district's future collection, so the observed schedule neither requires such a weight nor reveals it. The only number that remains free, the level of the district's total, was assigned to the office and varied with the district's current harvest and its share of failed farms (Section 4.1); it is not a choice over the future. And under Assumption 1 nothing the office did through a harvest figure could change what it collected in the following year (Appendix B). The office's problem therefore reduces to a single year, and it is written that way. This is a statement about what the accounts can and cannot reveal, not a claim that district officials disregarded the future; a specification in which the office weighs current against future collection was estimated and returned a weight above one, which is not admissible (Section 6.5).

What the model does not determine. The total T_D combined the norm, the reserve of obligations above the approved plan, and adjustments made during the campaign (Section 2), and the office observed the district's harvest as the campaign proceeded, so T_D may have depended on the district's harvest and was reduced where failure was certified (Section 4.1). Section 6.5 explains why its level is not identified. The failure threshold y_F is also taken as given; it was applied by the commission, not chosen by the office. The discretion that remained at the district lay in the level of the collection above the norm, the wedge of Section 4.1, and in the certification of failure on which the reduction to the statute depended, which the commission exercised. Neither gave an individual farm any influence over its own obligation. A farm could not change a district-wide multiple by misreporting its own harvest, and it could not obtain certification by a report, because the commission assessed the standing crop. Within the information restriction the office could differentiate among farms on characteristics it could verify and the farm could not alter, such as location, and the accounts show little sign that it did (Section 4.1).

5.5 The Farm

The collective farm sets no obligation. It enters the model through one decision, the division of the grain left to it after its obligations between seed for the next sowing and food for its members. This division is not derived from an optimisation problem. It is a rule estimated from the farms' own accounts. The model requires of the farm only the mapping from the grain left to it into the seed it holds in the spring, which the centre inspects, and into the food its members receive over the winter, which determines the outcomes of Section 5.7. Both are observed, and the rule that connects them is the same across farms.

The residual and its division. Farm i harvests H_i and pays the delivery x_i to the district office and, if it is served by a station, the fee M_i . What remains is the residual,

$$R_i = H_i - x_i - M_i. \quad (3)$$

The farm requires $\varsigma \bar{a}_i$ centners of seed to sow its registered area and is required to hold an insurance margin of κ above that requirement (Section 5.2). When the residual is large enough, the farm fills its seed fund to $(1 + \kappa)\varsigma \bar{a}_i$ and distributes the remainder to its members. When it is not, the farm puts a constant share φ of the residual into seed and distributes the rest:

$$s_i = \min\{(1 + \kappa)\varsigma \bar{a}_i, \varphi R_i\}, \quad c_i = R_i - s_i, \quad (4)$$

where s_i is the seed held in the spring and c_i the grain available for distribution. The seed shortfall is the excess of the requirement for the registered area over the seed held,

$$\sigma_i = \max\{\varsigma \bar{a}_i - s_i, 0\}. \quad (5)$$

A farm is short of seed, $\sigma_i > 0$, when $\varphi R_i < \varsigma \bar{a}_i$, that is, when its residual falls below $\varsigma \bar{a}_i / \varphi$. On such a farm every centner taken by an obligation reduces seed by φ centners and food by $1 - \varphi$ centners.

From seed to area. Seed determines the area a farm sows and not the yield per hectare: in the accounts, seed sown is proportional to area sown at the rate ς , and the change in a farm's area between the two years follows its loan and its shortfall with equal and opposite coefficients (Section 4.2). If the centre lends the farm r_i centners of seed before the sowing (Section 5.6), the area the farm sows in the following spring is

$$a'_i = \bar{a}_i - (\sigma_i - r_i) / \varsigma + g_f \bar{a}_i / 100, \quad (6)$$

the registered area less the part of the shortfall the loan did not cover, plus the land released from flax. A farm whose shortfall is filled returns to its registered area. This is the channel through which the grain a farm keeps in one year becomes its area, and hence the base of its obligations, in the next.

The evidence for the rule. The division is treated as a constant share rather than as a priority order or an optimising choice because that is the pattern in the accounts (Section 6.1). Among farms whose residual covered neither their seed requirement nor their members' food, seed and food both fell short in proportion; farms did not fill one need before starting on the other. The same division appears at a different level of aggregation: in the regional surveys of 1932 and 1933, the share of the gross harvest that Sverdlovsk farms put into seed and the share they distributed to members changed little between the two years, and by similar amounts. Seed, moreover, behaves as a technical requirement rather than as an input that farms vary: across the panel, seed sown is the seeding rate times the area sown, with little dispersion around that rate. A share ϕ below one means that a farm short of both needs sacrifices part of its next sowing to feed its members over the winter; this is the margin the centre's seed loan later restores.

What is held at its recorded value. The repayment of seed and food lent in earlier years, which ranked ahead of the seed fund in the statutory order of payments and took 1 per cent of the gross grain harvest in 1933 and 3 per cent in 1934, and the fodder fund, which ranked after the seed fund and took 8 and 12 per cent, are held at their recorded values and not modelled (Section 2). Because neither appears in (3), a given loan restores slightly less area than (6) implies; Section 6.3 states the direction in which this affects the bound on the centre's discount factor. The harvest H_i is likewise held at its recorded value: the model contains no response of the farm's effort or of its yield to the obligation it faces. Section 7.1 reports the test on which this rests, that the district's collection above the norm in 1933 did not predict the change in a farm's yield, area, output or labour days in 1934.

5.6 The Centre

Section 4.2 documented the seed relief. Before the 1934 sowing the state lent seed to four fifths of the farms that were short; the loan equalled the shortfall at the median farm; the farms left without a loan were those with the smallest shortfalls; and the loan followed the farm's own shortfall rather than its district's. The loan was the state's response to a uniform claim, one rate per hectare for every farm of a district, that did not vary with the harvest. The statutory obligation could not be reduced when a farm's harvest failed; the district office waived its collection above the statute on

the farms certified as failed, and nothing else in the year adjusted a farm's burden to the outcome of its harvest. The loan is modelled because it is the one instrument through which the state corrected, farm by farm, a claim that did not vary with the harvest, so that part of the cost of a claim that did not vary with the harvest was repaired there, and the timing of the repair can be measured: the loan restored the next sowing after the members had passed the winter on what the collection left. It is also the only decision of the centre in which a trade-off between the present and the future can be observed, because seed lent in the spring is grain the state gives up now in exchange for a sowing, and therefore for obligations, in the following year; the size of the loan relative to the shortfall is informative about how the centre weighed that future. And the rule the model derives for the loan enters every counterfactual of Section 7: the relief is held at that rule, and its timing, seed in the spring rather than food in the autumn, is one of the components of the human cost. The norm itself is the policy change studied throughout the paper and enters the model as a parameter (Section 5.2). The centre's problem is written as a dynamic programme, the form of its solution is derived, and the observed relief is shown to bound the weight the centre placed on the following year.

The collection on a farm as a function of its area. Let a denote the area a farm sows in a given year, which may fall short of its registered area $\bar{a}$ when the farm is short of seed. The delivery is computed from the registered area, and the following year's registered area follows the area last sown with response β (Section 5.2), so the base of the delivery on a farm that sows a is $\beta a + (1 - \beta)\bar{a}$. What the office collects on that base is its policy function of Proposition 1: the statute on a farm whose yield falls below the failure threshold, and the district multiple on every other farm. The station's fee is the share m of the harvest, and a farm cannot deliver more than it harvested. Writing y for the yield, the grain the centre collects from the farm in the year is

$$x(y, a) = \min\{ya, \zeta(y)n(\beta a + (1 - \beta)\bar{a}) + mya\}, \quad \zeta(y) = \mu_D \text{ if } y \geq y_F \text{ and } \zeta(y) = 1 \text{ if } y < y_F. \quad (7)$$

The centre's problem is therefore solved given the office's optimal collection, with the level of the district multiple taken from the accounts (Section 6.2). The collection includes the office's collection above the norm, because that grain also reached the state. The residual, the seed held and the seed shortfall follow from (3) to (5) with a in place of $\bar{a}$ in the harvest; in what follows $\sigma(y, a)$ denotes the shortfall net of the land released from flax, the quantity of seed the centre must send to return the farm to its registered area.

The centre's problem. The state of a farm at the start of a year is the area a it can sow. Let $v(a)$ denote the value of the farm to the centre: the expected present value of what the centre will collect

from it from this year on, net of the cost of the seed it will lend. After the harvest and the collection the centre observes the shortfall σ and chooses a loan r between nothing and the whole shortfall. A centner lent costs the centre $1 + \lambda$ centners (Section 5.2). The farm then sows $\bar{a} - (\sigma - r)/\varsigma$ in the following spring. The value function satisfies

$$v(a) = \mathbb{E}_y [x(y, a) + w(\sigma(y, a))], \quad w(\sigma) = \max_{0 \leq r \leq \sigma} \{-(1 + \lambda)r + \delta v(\bar{a} - (\sigma - r)/\varsigma)\}, \quad (8)$$

where the expectation is over the yield distribution of Section 5.2 and δ is the centre's weight on the following year. The loan can be written as the area a' to which the centre restores the farm. Substituting $r = \sigma - \varsigma(\bar{a} - a')$,

$$w(\sigma) = -(1 + \lambda)\sigma + (1 + \lambda)\varsigma\bar{a} + \max_{\bar{a} - \sigma/\varsigma \leq a' \leq \bar{a}} \{\delta v(a') - (1 + \lambda)\varsigma a'\}. \quad (9)$$

In this form the quantity maximised does not depend on how large the shortfall is; the shortfall only fixes the lowest area at which the farm can be left. This property gives the solution its form.

Proposition 2 (The relief rule). *Let $x(y, a)$ be bounded and non-decreasing in a . (i) Equation (8) has a unique bounded solution v , and v is non-decreasing. (ii) If $\delta v(a') - (1 + \lambda)\varsigma a'$ is single-peaked in a' with its peak at a^* , the optimal loan is*

$$r^*(\sigma) = \max\{0, \sigma - \sigma^*\}, \quad \sigma^* = \varsigma(\bar{a} - a^*) : \quad (10)$$

a farm whose shortfall is smaller than the tolerance σ^ receives nothing, and every farm whose shortfall is larger is restored to the same area a^* . (iii) The tolerance is zero, so that every shortfall is filled, if and only if*

$$\delta v'(\bar{a}) \geq (1 + \lambda)\varsigma, \quad (11)$$

and under full filling

$$v'(\bar{a}) = \mathbb{E}[x_a] + (1 + \lambda)\phi \mathbb{E}[(y - x_a)\chi], \quad (12)$$

where x_a is the derivative of (7) with respect to the area, the collection that one more sown hectare attracts, and χ equals one on farms that are short of seed.

The proof is in Appendix A. Part (i) holds because the operator on the right of (8) is a contraction with modulus δ . Part (ii) is the restocking rule of inventory theory (Arrow et al., 1951): the stock is sowable area, and the centre restores every farm below a threshold to the same level; it is consistent with the relief Section 4.2 found in the accounts. Part (iii) gives the condition under which the threshold is the registered area itself: the discounted value to the centre of the last hectare of the plan must cover the seed that sows it and the charge on that seed. Equation (12)

gives that value. One more hectare brings the centre the obligation it attracts, and on a farm that is short of seed it also brings the share ϕ of the hectare's residual that the farm puts into seed, which is relief the centre no longer has to send.

The role of the base. Equation (12) shows why the response β matters. If the base of the delivery did not follow the area a farm sowed, so that $\beta = 0$, an additional hectare would attract only the fee, $v'(\bar{a})$ would be small, and condition (11) could hold only for a weight above one: no discount factor would justify sending seed at all. The relief is worth its cost to the centre because the obligation follows the area with a lag. The rule that raised a farm's obligation when its recorded area recovered is the same rule that made its recovery worth financing.

What the observed relief identifies. A positive tolerance would identify δ as a point, since σ^* in (10) is a function of δ . The tolerance of zero that Section 4.2 shows identifies only the bound (11): when every shortfall is filled, the accounts imply that δ is at least $(1 + \lambda)\zeta/v'(\bar{a})$ and are silent about how much larger it is. Section 6.3 computes $v'(\bar{a})$ on the 1933 yield distribution, in which the shortfalls opened, and reports the bound with its sampling interval. The bound rests on one harvest, since 1934 produced few shortfalls and little relief, and the repayment of earlier loans and the fodder fund, held at their recorded values (Section 5.5), mean that a given loan restored slightly less area than (6) credits, so the bound as computed is slightly too low.

5.7 Outcomes

The model's outcomes are the food available to a farm's members and the number of members a year later. Both are computed from the grain that remains to the farm after the delivery and the station's fee, through the division rule of Section 5.5 and the population response estimated in Section 4.3. Population is an outcome of the model and not a state variable of any agent's problem: the centre's problem is written on collection and on the area a farm can sow, which is what the centre observed and acted on, and the members' food had no register (Section 5.3). The human cost of any delivery rule is therefore computed after the model has been solved, from the food the rule leaves to the members.

Food per member. Let K_i denote the kilocalories a farm distributed to its members over the year from its grain, potatoes, vegetables and milk, constructed as described in Section 3, and N_i the number of its members. Food per member per day is

$$k_i = K_i / (365 N_i). \quad (13)$$

Under a counterfactual delivery that leaves the farm more or less grain than the observed one, food changes by the difference in grain, converted to kilocalories and multiplied by the pass-through π of Section 5.2. If the observed delivery is D_i and the counterfactual delivery is x'_i ,

$$K'_i = K_i + \pi \cdot 3,400 \cdot 100 \cdot (D_i - x'_i). \quad (14)$$

The same expression applies to a change in the station's fee and to a seed loan counted as food rather than as seed. Food per member is bounded below at 300 kilocalories per member per day before the response is applied, so that a counterfactual delivery that exceeds a farm's harvest produces a bounded response.

Members a year later. The number of members a farm has a year later responds to food per member with the constant elasticity ε of Section 4.3:

$$N'_i = N_i (k'_i/k_i)^\varepsilon, \quad (15)$$

where k_i is the observed food per member and k'_i the food per member under the counterfactual delivery. The shares of the response accounted for by deaths and by departures, also estimated in Section 4.3, are applied to every counterfactual. For the 1933 harvest the members a year later are observed. For the 1934 harvest the response is applied to a year the panel does not observe, and the results for 1934 are marked as extrapolations wherever they appear.

The population cost of a change in deliveries. The number of people lost when the deliveries of a district's farms are changed from x^0 to x^1 is the difference in members a year later:

$$L = \sum_i N_i [(k_i^0/k_i)^\varepsilon - (k_i^1/k_i)^\varepsilon], \quad (16)$$

positive when the second set of deliveries leaves less food. Because the response is a power function of food, it is never zero and it is concave: a given quantity of grain taken from a farm costs more members where food per member is already low. The same total extracted from the farms of a district therefore costs more people when it falls on farms whose food is near or below the subsistence line of 2,100 kilocalories per member per day than when it falls on farms above it. This is the sense in which the paper's subject is subsistence agriculture. The line itself enters the results as a descriptive count, the number of farm-years left below it under each delivery rule; it plays no part in the response, which is continuous in food.

What is held fixed. The harvest, the fee, the division rule and the population response are held at their observed or estimated values under every counterfactual; only the delivery changes (Section 5.5 for the harvest). The division rule is applied to the counterfactual residual as it was to the observed one, so a delivery that leaves a farm more grain leaves it more seed as well as more food, in the proportion the rule gives.

6 Estimation

The parameters of the model come from four sources: the statutes, the farm accounts, estimation on the panel, and, for the centre's discount factor, a lower bound derived from the observed seed relief. Section 6.1 gives the value and source of each. Section 6.2 checks the solution of the office's problem against the accounts. Proposition 1 says that the office collected the statutory amount and no more from farms certified as failed; the farms so certified in 1933 allow this to be tested. The same subsection reports the level of the district totals, which the model takes as given. Section 6.3 solves the centre's problem on the 1933 yield distribution, reports the value of a restored hectare, and derives the bound on the discount factor with its sampling interval. Section 6.4 sets the model's predictions beside the facts of Section 4 in one table. Section 6.5 states what the accounts do not identify.

6.1 Parameters

Table 7 lists the parameters of the model in four groups: those fixed by statute, those read from the accounts, those estimated from the panel, and those calibrated or bounded.

The statutes fix the norm, the charge on seed lent, and the band within which the insurance margin on the seed fund lies. The norm is applied to each farm's sown area by crop at the published rates for farms served or not served by a machine-tractor station; at the median farm the statutory obligation is 1.8 centners per hectare, and a farm served by a station owed about four fifths of what an unserved farm with the same crops owed. The charge of ten poods on every hundred lent gives $\lambda = 0.10$. The insurance margin is set at 0.11, inside the statutory band of 10 to 15 per cent; it is the median excess of the seed fund over the requirement, as a share of the requirement, on the farms whose residual exceeded both their needs by at least a quarter.

From the accounts come the registered area of each farm; the seeding rate, seed sown over area sown, 1.52 centners per hectare with little variation across farms; the machine-tractor station's fee in kind as a share of the gross harvest, 0.086 at the median of the farms that paid one; the two multiples of the statute for each district and year (Section 4.1); and the failure threshold of four centners per hectare, at which the deliveries break (Section 4.1).

The yield process, the division share, the response of the next year's base to a restored hectare, the land released from flax and the population response are estimated from the panel. Log yield net of district-year effects has a standard deviation of 0.25, and its correlation of 0.52 between the two years divides that dispersion into a permanent farm component and a transitory component of about equal size. The division share, 0.57, is the median share of the residual that farms put into seed among the 127 farms of 1933 whose residual covered neither their seed requirement nor the grain needed to keep their members at the subsistence line; the median over all 244 farms with an observed division, 0.48, is used as an alternative value in Section 6.3. The response of the next year's base to a restored hectare, 0.86, is the elasticity of a district's 1934 collective sowing plan to its 1932 recorded sown area across the 31 agricultural districts. The land released from flax is 6.27 hectares per hundred hectares of the grain plan on the 244 farms with a grain area in both years. The population response is the elasticity of Section 4.3.

The pass-through of grain into food and the subsistence line are calibrated. When a counterfactual delivery leaves a farm more grain than it actually kept, the model must specify how much of that grain would have been eaten by the members and how much put into seed and fodder. In the main results all of it is eaten, $\pi = 1$. Section 7.4 repeats the results with 60 per cent eaten, and with the share that the farm's own division rule leaves for food when seed is short, $1 - \phi$. The subsistence line is 2,100 kilocalories per member per day (Section 3). The lower bound on the centre's discount factor is derived in Section 6.3.

6.2 The District Office

The solution of the office's problem in Section 5.4 has two testable implications and one free number. The implications are that all farms in a district not certified as failed deliver one common multiple of the statute, and that farms certified as failed deliver the statute and no more. The free number is the district's total T_D , which the model does not explain. The second implication is tested below on the farms of 1933, the first having been documented in Section 4.1; the level of the totals is then read from the accounts, and its consistency with the response of the totals to the harvest is checked. Table 8 reports the estimates.

Failed farms delivered the statute. On the 45 farms of 1933 with a barn yield below four centners per hectare, the multiple of the statute is 0.98 at the median, with quartiles 0.85 and 1.11; pooled over these farms, deliveries were 0.97 of the statute. On the 33 farms with yields between three and four centners the median is 1.03. Below three centners the statute exceeded what the farm could deliver, and the shortfall went into arrears (Section 2). The model ties the reduction to the commission's certificate, not to the farm's own harvest, and the two can be separated on served

Table 7: Parameters of the model.

A. Fixed by statute			
Symbol	Definition	Value	Source
n_i	delivery norm per registered hectare, by crop and by whether a machine-tractor station served the farm	1.807 centners per hectare at the median farm; the rate for a served farm is 0.79 of the rate for an unserved farm with the same crops	decree of 27 January 1933, applied to each farm's sown area by crop; 493 farm-years
λ	charge on seed lent	0.10	decision of 18 February 1933 statutory band of 10 to 15 per cent; the median fund of the 56 farms whose residual exceeded both needs by at least a quarter, 1933
κ	insurance margin on the seed fund	0.11, with quartiles 0.04 and 0.17	
B. Read from the accounts			
Symbol	Definition	Value	Sample
$\bar{a}_i$	registered grain area	394 hectares at the median farm	panel, 1933
ς	seed required per hectare sown	1.52 centners per hectare	seed sown over area sown; coefficient of variation across farms 0.12
m	station's fee in kind as a share of the gross harvest	0.086	median of fee over gross harvest on the 225 farm-years that paid a fee; 0.085 on the 97 of 1933
μ_D, ζ_D	multiples of the statute collected from farms not certified as failed and from failed farms	one pair for each district and year	70 district-years with at least three farms, 36 districts (Section 4.1)
y_F	yield below which a farm was certified as failed	4 centners per hectare	the break in the multiple of the statute (Section 4.1)
C. Estimated from the panel			
Symbol	Definition	Value	Sample
y_i	yield process: standard deviation of log yield net of district-year effects; correlation across years; permanent and transitory standard deviations	0.25; 0.52; 0.18 and 0.17	489 farm-years in 86 district-year cells, 245 farms; 244 farms in both years
φ	share of the residual put into seed when both needs are unmet	0.57, with quartiles 0.45 and 0.63; 0.48 over all farms	127 farms short of both needs, 1933; 244 farms with an observed division
β	response of the next year's base to a restored hectare	0.86 (R^2 0.90)	log 1934 sowing plan on log 1932 sown area, 31 agricultural districts
g_f	land released from flax per hundred hectares of the grain plan	6.27 hectares	244 farms with a grain area in both years
ε	elasticity of members a year later to food per member	0.135 (standard error 0.032); deaths 37 per cent and departures 63 per cent of the response	252 farms (Section 4.3)
D. Calibrated or bounded			
Symbol	Definition	Value	Source
π	share of a centner not delivered that is eaten by the members	1 in 37 in main results; 0.6 and $1 - \varphi$ in Section 7.4	calibrated
—	subsistence line	2,100 kilocalories per member per day	Section 3
β	response of the next year's base to a restored hectare	0.86 (R^2 0.90)	log 1934 sowing plan on log 1932 sown area, 31 agricultural districts

farms, whose assessed harvest is recoverable from the fee: of the 19 served farms of 1933 with a barn yield below four centners, the 11 that the commission had assessed below four delivered the statute, a median multiple of 1.02, and the 8 that it had assessed at four or above delivered a median of 1.22, a fifth above the statute, like the farms that were not certified. Both forms of the prediction are consistent with the accounts.

The level of the totals. The totals are read from the accounts as the multiple of the statute in each district and year. In 1933 the upper level of the district multiples, measured as their 95th percentile, was 1.55, and in 1934 it was 1.36, with 90 per cent intervals of 1.50 to 1.56 and 1.34 to 1.39 from a bootstrap that resamples districts; about four fifths of the district-years of 1933 and nine tenths of those of 1934 lie at that level. The districts below it are the districts with failed farms: in 1933 the distance of a district's multiple below the upper level rises with the share of its farms that had failed, by about half the statute over the range from no failed farms to all, and the share of its farms below the subsistence line adds nothing once the failed share and the harvest are held fixed (Table 8). At the median district of 1933 the farms that had not failed harvested 3.5 times their statutory obligation and delivered 1.3 times it.

Consistency of the level with the response. A schedule in which most districts sit at an upper bound and only the districts below it move with their harvest implies that a change in a district's harvest between the two years should move its multiple by a fifth to a quarter of the statute per log point. The response estimated directly on the change between the two years in Section 4.1 is a quarter. A schedule without an upper bound would imply a response about five times as large.

Alternatives rejected. An office that set its own total to fill a fixed plan with a safety margin, spreading expected failures over the farms that could pay, would collect multiples between 1.04 and 1.17 and collect more from the remaining farms where more farms had failed; the observed multiples run from 0.91 to 1.56 and fall with the failed share. An office that valued the following year's collection as the centre does would have collected up to the district's physical capacity, since the seed it removed from a farm would be replaced by the centre at a charge (Section 5.6); the observed collection, a third of the harvest of the farms that had not failed, is far below capacity.

Table 8: The multiple on failed farms by barn yield and by the commission's assessment; the upper level by year with its interval and the share of district-years at it; the distance below the upper level on the failed share and on the share of farms below subsistence; and the response implied by the level beside the response estimated on the change between the two years.

to be built from the run outputs

Figure to be drawn: the district-year multiple against the ratio of the district's harvest to its statutory obligation, by year, with the upper level marked.

Figure 7: The district-year multiple against the ratio of the district's harvest to its statutory obligation, by year, with the upper level marked.

6.3 The Centre

The centre's problem of Section 5.6 is solved numerically. The solution gives the value to the centre of a restored hectare and, together with the observation that every seed shortfall was filled, the lower bound on the centre's discount factor. The problem is solved for a farm with a plan of 100 hectares on the yield distribution of 1933, the harvest in which the shortfalls opened, with the parameters of Table 7; the collection on the farm is the office's policy function of Proposition 1, the statute on draws below the failure threshold and the district multiple on the others, applied to the norm and the response of the base to the area sown, plus the station's fee at the average share. The value function is computed by iterating equation (8) on a grid of 501 areas and 400 yield draws until it converges. Table 9 reports the results, and Figure 8 plots the marginal value of a hectare and the tolerance of the relief rule against the discount factor.

The value of a restored hectare. The marginal value of a hectare to the centre, $v'(a)$, falls as the farm's area rises: at a discount factor of one half it is 4.5 centners of grain a year at 20 hectares of the 100-hectare plan, 3.9 at 60 and 2.9 at the plan itself. A hectare restored to a farm that can sow only part of its plan is worth more than a hectare at the plan because the farm is more likely to be short of seed, so the hectare saves relief as well as attracting an obligation. The concavity of the value function is what makes the relief a restocking rule (Proposition 2) and what makes the office prefer to spare failed farms (Proposition 1).

The bound on the discount factor. By Proposition 2 the centre fills every shortfall if and only if $\delta v'(\bar{a})$ is at least $(1 + \lambda)\zeta$, the seed that sows the last hectare of the plan together with the charge on it, 1.67 centners. On the 1933 distribution $v'(\bar{a})$ is 2.88 centners, so the centre's discount factor is at least 0.58, with a 90 per cent interval from 0.55 to 0.59 obtained by resampling farms, both years of a farm together, and recomputing the yield distribution, the norm, the fee share and the released flax land in each of 1,000 draws. The bound lies between 0.50 and 0.62 across the alternative configurations reported in Table 9: the district multiple, the norm and the fee share taken from 1933 alone rather than from both years; the division share at its full-sample value; the pooled two-year yield distribution; and readings of the yield process that treat the permanent component of a farm's yield differently. The numerical solution agrees with the closed form: on the grid, every shortfall is filled from a discount factor of 0.60 upwards, and below it the centre leaves the smallest shortfalls unfilled.

Fit to the frequency of shortage. On the 1933 distribution the model's frequency of seed shortage matches the panel's. A farm at its plan is short of seed with probability 0.41 in the model, against 0.39 among the panel's farms at their registered area and 0.33 net of the land released from flax. On the pooled two-year distribution the model's probability is 0.17, and at the 1934 level of yields 0.05, which is consistent with the few shortfalls and the little relief of the 1934 harvest (Section 4.2) and is the reason the bound is computed on 1933.

What the bound rests on. The bound depends on the response of the base to the area sown, on the division share and on the transition from seed to area. If the base of the delivery did not follow the area a farm sowed, a restored hectare would attract only the fee, and the bound would exceed one under every reading of the yield process, so that no discount factor would rationalise sending seed at all (Section 5.6). Through the division share a restored hectare saves relief on farms that are short of seed. The transition from seed to area is taken from the accounting identity that seed sown is the seeding rate times the area sown, one hectare per 1.52 centners, and not from the estimated conversion of Section 4.2, which is attenuated by measurement error in the loan and the shortfall. The bound is a bound and not a point, because filled shortfalls are silent about how much more than the minimum the centre valued the next harvest; and the repayment of earlier loans and the fodder fund, held at their recorded values (Section 5.5), mean that a loan restored slightly less area than that identity credits, so the true bound is slightly higher than the one reported.

Table 9: The bound on the discount factor and its interval under each configuration, the value of a hectare at the plan, and the model's probability of shortage beside the panel's.

to be built from the run outputs

Figure to be drawn: the marginal value of a hectare against the area sown, and the tolerance of the relief rule against the discount factor, on the 1933 distribution.

Figure 8: The marginal value of a hectare against the area sown, and the tolerance of the relief rule against the discount factor, on the 1933 distribution.

6.4 Fit

Table 10 sets each prediction of the model beside the corresponding fact and names the subsection in which the fact is reported. The predictions are of three kinds: those that follow from the assumptions of Section 5.1, those that follow from the office's problem, and those that follow from the centre's problem. Every fact in the table is reported in Section 4 or in Sections 6.2 and 6.3, and the table adds no estimate to them.

The level at which failed farms delivered varies across districts and rises with the district's multiple; this is consistent with the model only if the district norms set under the law differed from the regional rate, which the accounts cannot confirm (Section 6.5). Some farms received seed although their fund already covered their registered area, because the area they sowed in 1934 exceeded it; such loans lie outside the relief rule, which restores the registered area and does not finance sowing beyond it.

6.5 Identification

The accounts identify the form of the two rules, the parameters of Table 7 and the bound on the centre's discount factor. They do not identify the level of the district totals, the office's payoff parameters, the centre's discount factor above the bound, the size of the misreporting under the obligations that stayed on the harvest, or the grain that a claim on the harvest would have extracted. Table 11 lists these objects with the reason and the treatment adopted.

A district's total combined the norm with the reserve of obligations handed down above the

Table 10: Predictions of the model and the corresponding facts.

Prediction	Fact in the accounts	Reported in
<i>From the assumptions</i>		
Within a district the delivery does not vary with the farm's own harvest	elasticity of the multiple to the farm's own yield is zero in both years; twelve farm characteristics explain six per cent of the variation within districts	4.1
A served and an unserved farm in the same district deliver the same multiple of their own statutes	difference of 0.013 log points, standard error 0.021	4.1
The two obligations that stayed on the harvest show signs of misreporting	the station's fee rises with the farm's harvest and is computed on an assessment above the barn harvest; the flax crop recorded in 1933 is a fraction of that of 1934	Appendix B
The collection above the norm did not change the following harvest	the district's 1933 multiple does not predict the change in a farm's yield, area, output or labour days in 1934	7.1
<i>From the office's problem</i>		
Farms certified as failed deliver the statute and no more	median multiple 0.98 on the failed farms of 1933; farms the commission assessed below the threshold deliver the statute, farms it assessed above it deliver the district multiple	6.2
All other farms in a district deliver one common multiple	one level per district-year on non-failed farms	4.1
The failed level is one number per district	its variance within districts is the same share of the total as for the other farms	4.1
The number of farms in a district does not affect the multiple	coefficient on the number of farms indistinguishable from zero	4.1
Where the total was reduced for certified failure, the multiple falls with the failed share	the multiple falls with the failed share in levels and in changes between the two years	4.1
The level of the totals and their response to the harvest are consistent	the level implies a response of a fifth to a quarter of the statute per log point; the estimated response is a quarter	6.2
<i>From the centre's problem</i>		
The loan equals the shortfall	ratio of loan to shortfall 1.03 at the median farm, 1.05 in the aggregate	4.2
The loan depends on the farm's own shortfall and not on its district's	within districts the loan rises 0.78 per unit of the farm's own shortfall; the neighbours' shortfalls enter at 0.07 (0.11)	4.2
The farms left unrelieved are those with the smallest shortfalls	share relieved rises from two thirds in the lowest quartile of the shortfall to nine tenths in the third	4.2
The rule applies to grain seed against the shortfall and not to seed in general	the flax seed sold in the same weeks follows the flax area and not the grain shortfall	4.2
A loan equal to the shortfall leaves the area sown unchanged	the loan and the shortfall enter the change in area with equal and opposite coefficients	4.2
The frequency of seed shortage at the registered area	0.41 in the model on the 1933 distribution; 0.39 in the panel, 0.33 net of released flax land	6.3
The division of the residual is a constant share, not a priority order	on the 127 farms short of both needs, seed and food fell short in proportion	6.1

approved plan, from which the reductions granted during the campaign to districts whose crop had failed were financed (Section 2). The district norms set under the law are not among the sources available to me. The accounts therefore show what each district collected and how the collection varied with the harvest and the failed share, but not how the level was set, and the multiple of the statute cannot be split into the district's own norm and the office's collection above it. On the thirteen districts where the level on failed farms and the level on the other farms are both observed, the office's margin above the district norm would be about a tenth, but the estimate is unstable and is not used.

The office's gain per centner, its penalties and its weight on the future leave no trace in the schedule, because both of its decisions are corner solutions (Section 5.4). The weight was nonetheless estimated from the marginal condition that equates the value to the office of a centner collected with the future collection that the centner removes, the latter computed as the office could have expected it at the time of collection, from the 1933 yield distribution and the value function of Section 6.3. On the 34 district-years of 1933 the condition holds only at a weight of 1.74, with a 90 per cent interval from 1.53 to 2.09. A weight above one is inadmissible: at any admissible weight the expected future loss from a centner collected is too small to have stopped the office at the multiple observed, so an office that made this trade-off would have collected more than it did. The level of the multiple was therefore determined by something other than this trade-off, which is the reason the office's problem is written as a corner at an assigned total. A penalty for a shortfall of the total cannot be identified either, because at the observed multiples the probability of a shortfall was negligible.

7 Counterfactuals

The rules of Sections 5 and 6 convert any schedule of deliveries into food per member and members a year later. They are used here to measure what the norm cost in people and to divide the cost between the information constraint and the incentive constraint. Section 7.1 defines the counterfactual schedules, states what each holds fixed and what it changes, and gives the design of the decomposition. Section 7.2 reports the human cost of the norm as run. Section 7.3 divides it into the uniformity of the norm across farms, the level of the wedge, the misallocation of the wedge, and the timing of the seed relief. Section 7.4 reports the robustness of the results to the pass-through of grain into food, to the choice of the informed allocation, and to the year.

Table 11: Objects not identified from the accounts.

Object	Reason	Treatment
the level of a district's total	the norm, the reserve above the approved plan and the reductions financed from it combined; the district norms not available	reported as observed; its response to the harvest estimated on the change between the two years
the split of the multiple into the district's norm and the office's margin	the district norms not available	the multiple measured against the regional rate
the office's gain, penalties and weight on the future	both decisions are corners; the marginal condition holds only at a weight above one	the office's problem written as a corner at an assigned total
the centre's discount factor above the bound	filled shortfalls are silent above the threshold	the bound reported with its interval
the assessor's overstatement and the farm's understatement under the fee	inseparable within the difference between the assessed and the barn harvest	the difference reported as a whole (Appendix B)
concealment and withdrawal of effort under the flax contract	inseparable in the recorded crop	aggregates reported only (Appendix B)
the grain a claim on the harvest would have extracted	requires the reporting game solved	totals held at observed values in every counterfactual
the reduced norms of 1934	the annex to the decree is not among the sources available to me	the difference between the years read as the change in the statute and in the plan together
the waiver on failed farms in 1934	only three farms certified as failed in 1934	not measured
the population response after the 1934 harvest	the members of 1935 not observed	the 1933 response applied, marked as an extrapolation

7.1 Design

Every counterfactual is a schedule of deliveries substituted for the observed one, farm by farm, with the harvest, the station's fee, the farm's division rule and the population response held at their observed or estimated values. Each schedule changes one element of the observed rules and leaves the rest as they were. The total collected in a district is not predicted by the model: every schedule collects either the statutory total of the district or the total the district office actually collected, and the schedules differ only in which farms bear it. What is measured is therefore the cost of allocation and timing under the norm, not the cost of the norm relative to a different level of extraction, and not the cost relative to the contract system, whose strategic equilibrium the accounts do not identify (Section 6.5).

The schedules. Table 12 lists them. The statute alone, (a), is the norm as written: each farm delivers its statutory obligation and no more. The district multiple, (b), is the office's schedule as observed: the statute on farms certified as failed and the district multiple on the rest; it is the policy as run. Schedule (c) is (b) with the seed loan of the following spring counted as food in the autumn instead, on the farms that received it; it measures the timing of the relief. Schedules (d) and (e) hold the office's total and allocate it with information the office did not have: (d) in proportion to each farm's harvest, and (e) so that every farm the total allows is held at the subsistence line, the remainder being allocated in proportion to the harvest. Schedules (g) and (f) do the same at the statutory total, (g) in proportion to the harvest and (f) with subsistence first; (f) is the benchmark against which every cost is measured. A second informed allocation, the concentrated allocation, pushes a district's shortfall onto as few farms as possible, so that the number of farms below the line is the smallest achievable; it is reported as a bound on that number and not as a welfare measure, because the population response is concave and a shortfall concentrated on few farms costs more people than the same shortfall spread over many.

The samples. The schedules are computed on the 440 farm-years in the 70 district-years with at least three farms, 213 farm-years in 1933 and 227 in 1934, holding 124,864 members. A schedule that holds farms at the subsistence line can do so only where the district's total allows: in 21 of the 70 district-years under the office's total, and in 14 under the statutory total, the total exceeds what the district's farms can deliver while every farm stays at the line, and the remainder is then allocated in proportion to the harvest. Two farm-years cannot reach the line even at a zero delivery.

The decomposition. The schedules form a two-by-two design: the total collected, statutory or the office's, against the allocation, uniform per hectare as the law wrote it or on each farm's harvest with subsistence first. People lost relative to the benchmark, the statutory total allocated on harvests

(f), are read down a column and along a row. Down a column, from the statutory total to the office's total at a given allocation, is the cost of the wedge, the offices' collection above the norm, which the model attributes to the incentive constraint at the district. Along a row, from the informed allocation to the uniform one at a given total, is the cost of allocating per hectare rather than on harvests, the information constraint: at the statutory total it is the uniformity of the norm, at the office's total it is the misallocation of the wedge. The timing of the relief is a separate comparison, (b) against (c). The station's fee enters once, as the fee at the statutory rate on the barn harvest in place of the fee on the assessed harvest, on the farms where the assessed harvest is recoverable.

The harvest held fixed. Holding the harvest at its recorded value under every schedule assumes that the collection did not change what the farms produced. This is tested on the 176 farms observed in both years with a district multiple computed from the other farms of their district: the district's 1933 multiple does not predict the change in a farm's yield between 1933 and 1934, with the farm's 1933 yield and area held fixed, nor the change in its sown area, its output or its labour days per hectare (Table 12, notes). Across the range of multiples observed, the interval on the yield coefficient rules out an effect larger than about a tenth of the yield. The collection above the norm re-allocated the harvest between the state and the farm, and there is no sign that it reduced the harvest that followed.

Conversion to people. Under each schedule the grain left to a farm is converted to food per member by equation (14) and to members a year later by equation (15); the people lost between two schedules are given by equation (16), and the division into deaths and departures follows the shares of Section 4.3. The intervals reported in Sections 7.2 and 7.3 carry the standard error of the population response through the same equations and span 1.645 standard errors on either side of the estimate. Results for the 1934 harvest apply the response estimated on 1933 to a year the panel does not observe.

7.2 The Human Cost

The policy as run cost 4,392 people on the panel's farms over the two harvests, one member in twenty-eight, relative to a claim of the same statutory total allocated on the farms' harvests with subsistence first. Table 13 reports the four schedules of the two-by-two design with the number of farm-years each leaves below the subsistence line and the people it costs, in total and by year.

Under the policy as run, 161 of the 440 farm-years were left below 2,100 kilocalories per member per day; under the benchmark, the statutory total allocated by a party that saw each farm's harvest, 90 would have been. The difference in members a year later between the two schedules

Table 12: The counterfactual schedules.

Schedule	Delivery of farm i	Total collected	What it represents
(a) the statute	$n_i \bar{a}_i$	statutory	the norm as written
(b) the district multiple	$n_i \bar{a}_i$ times the district multiple, or the statute if certified as failed	the office's	the policy as run
(c) (b) net of the loan	(b) less the seed loan received in the spring	the office's, less the loan	the relief received as food in the autumn
(d) proportional, office's total	the office's total in proportion to the farm's harvest	the office's	the office's total allocated on harvests
(e) subsistence first, office's total	every farm held at 2,100 kcal per member per day where the total allows, the remainder in proportion to the harvest	the office's	the office's total allocated by a party that sees harvests
(f) subsistence first, statutory total	as (e) at the statutory total	statutory	the benchmark
(g) proportional, statutory total	as (d) at the statutory total	statutory	the statutory total allocated on harvests
(e'), (f') concentrated	a district's shortfall concentrated on the fewest farms	as (e), (f)	bounds on the number of farms below the line, not welfare measures

Notes. 440 farm-years in 70 district-years with at least three farms; 124,864 members. Held fixed under every schedule: the harvest, the station's fee, the division rule, the population response. The 1934 rows extrapolate the response estimated on 1933. Test of the harvest held fixed: the change in log yield between 1933 and 1934 on the district's 1933 multiple computed from the other farms of the district, with 1933 log yield and log area held fixed, coefficient -0.01 , standard error 0.15 , 176 farms; the same regressions for sown area, output and labour days per hectare give coefficients indistinguishable from zero.

is 4,392 people, two thirds of them for the 1933 harvest; about 1,600 are deaths and about 2,800 departures, in the proportions of Section 4.3. The interval implied by the standard error of the population response runs from about 2,600 to 6,200 people.

The cost concentrates where the claim crossed the subsistence line. In 1933, at a median yield of 5.6 centners per hectare, the statute alone left 32 per cent of farm-years below the line and the deliveries actually collected 58 per cent; in 1934, with a harvest three fifths larger under the same rule, the shares were 10 and 16 per cent, and the cost of the same rule fell by half. The population response is continuous in food and never zero, so the concentration is a matter of degree and not of a threshold: a centner taken from a farm already near the line costs more members than the same centner taken from a farm above it.

The statute alone, without any collection above it, would have left 92 farm-years below the line and cost 707 people relative to the benchmark; the rest of the cost, 3,685 people, is the cost of the wedge, the offices' collection above the norm. Section 7.3 divides the total along these lines.

Table 13: The human cost of the norm as run.

Schedule	Farm- years below 2,100 kcal (of 440)	Below the line, 1933 (of 213)	Below the line, 1934 (of 227)	People lost relative to the bench- mark	of which 1933	of which 1934
(f) statutory total, subsistence first (benchmark)	90	75	15	0	0	0
(a) the statute, uniform per hectare	92	69	23	707	471	237
(e) the office's total, subsistence first	138	123	15	3,463	2,293	1,171
(b) the office's total, uniform per hectare: the policy as run	161	124	37	4,392	2,911	1,482
observed deliveries	167	134	33	4,247	2,803	1,445

Notes. People lost are the difference in members a year later between the schedule and the benchmark, from the response of Section 4.3 at a pass-through of one; deaths are 37 per cent of each figure and departures 63 per cent. The interval on the total from the standard error of the response is about 2,600 to 6,200. The 1934 figures apply the response estimated on 1933 to a year the panel does not observe. The observed row differs from (b) because a few farms delivered above or below their district's multiple.

7.3 Decomposition

The 4,392 people divide along the two-by-two design of Section 7.1 into the cost of the uniform claim, the cost of the misallocation of the wedge and the cost of its level; the timing of the seed relief is measured by a separate comparison. Table 14 reports each part with the counterfactual that defines it, the constraint it measures, and the grain involved.

The uniform claim: 707 people. Moving from the benchmark, the statutory total allocated on harvests with subsistence first, to the statute as written, the same total uniform per hectare, costs 707 people, with an interval from 417 to 1,019. This is the cost of the norm's own form: one rate per hectare for every farm of the district, fixed before the sowing, so that a farm whose crop failed owed the same per hectare as a farm whose crop did not. It is the cost of the information the centre gave up when it wrote the claim on the base it could verify. The grain moved between farms by this reallocation is 33,313 centners, nine per cent of the statutory total.

The wedge: 3,685 people. Moving from the statute to the deliveries the offices actually collected, at the office's two-level schedule, costs 3,685 people, with an interval from 2,224 to 5,179; 2,440 of them for the 1933 harvest and 1,245 for the 1934 harvest. The grain is 82,838 centners, 18 per cent of the deliveries collected. This is the cost of the incentive constraint at the district: an office answerable for a district's total under criminal liability collected above a norm that the law had declared unchangeable, and the wedge fell where the norm already crossed the subsistence line.

Of which the misallocation of the wedge: 930 people. The wedge would have cost fewer people had the office been able to see which farms could bear it. Allocating the office's own total on the farms' harvests with subsistence first, instead of uniform per hectare, preserves 930 people, with an interval from 547 to 1,340; the grain moved between farms is 39,374 centners. This part of the cost of the wedge is attributed to the office's information constraint: one multiple for every farm it could not tell apart. The remainder, 2,755 people, is what the same wedge would have cost had it been allocated with full information; it is the cost of the level of the wedge, attributed to the incentive constraint.

The timing of the relief: 964 people. Before the 1934 sowing the state lent 13,546 centners of seed to 93 of the farm-years in the sample, 6 per cent of the 1933 deliveries and 29 per cent of the wedge of that year, at a charge of ten poods on every hundred, 1,355 centners. The loan restored the sowing; it did not restore the food of the winter that preceded it. Had the same grain been left on the same farms as food in the autumn, 19 more farm-years would have stayed above the subsistence line and 964 people would have been kept, the difference between schedules (b) and (c). This is the cost of the information the centre lacked in the autumn and obtained only in the late winter, in the one form in which it could obtain it, the seed stock against the register. Under an informed allocation of the office's total a quarter of the loan, 3,522 centners, would not have been needed, because the farms that received it would have kept the grain in the first place.

The station's fee, for comparison. On the 37 farms of 1933 whose fee was computed from an assessed harvest and whose assessment can be recovered, the fee exceeded the fee at the statutory rate on the barn harvest by 3,551 centners, 37 per cent of the statutory bill, at a cost of 216 people. This is the cost of the obligation that remained computed from an assessed harvest, on the farms where it can be measured; it is not part of the 4,392, which concern the delivery alone.

The components are the costs of the two constraints at the two levels at which the model locates them, given the norm. The uniformity of the norm, the misallocation of the wedge and the timing of the relief are the information cost of writing the claim on land: 707 and 930 people in allocation and 964 in timing. The level of the wedge, 2,755 people, is the cost of the incentive constraint. The station's fee measures the cost of the alternative base on the farms where it was kept; the flax contract, examined in Appendix B, shows only its direction.

Table 14: The components of the cost.

Component	Counterfactual	Whose constraint	Grain	People
the uniform claim	(f) → (a): the statutory total on land instead of on harvests	the centre's information constraint	33,313 c moved between farms	707 [417; 1,019]
the wedge, the offices' collection above the norm	(a) → (b)	the office's incentive constraint	82,838 c, 18 per cent of the deliveries	3,685 [2,224; 5,179]; 2,440 in 1933, 1,245 in 1934
of which its misallocation	(e) → (b): the same wedge on harvests with subsistence first	the office's information constraint	39,374 c moved between farms	930 [547; 1,340]
of which its level	the wedge allocated with full information	the office's incentive constraint	—	2,755
the timing of the relief	(b) → (c): the seed loan as autumn food	the centre's information constraint on the repair	13,546 c lent; charge 1,355 c; 3,522 c unnecessary under (e)	964
the station's fee on an assessed harvest	the fee at the statutory rate on the barn harvest, 37 farms	the incentive constraint under the crop share	3,551 c, 37 per cent of the bill	216

Notes. Intervals are ± 1.645 standard errors of the population response. The three parts of the delivery's cost sum to 4,392: 707 + 930 + 2,755; the timing of the relief is a separate comparison against the policy as run. The fee row is on the farms where the assessed harvest is recoverable and is not part of the total.

7.4 Robustness

The share of a centner not delivered that would have been eaten, the rule by which an informed party allocates a fixed total, and the inclusion of the 1934 harvest, for which the population re-

sponse is extrapolated, are not determined by the data. Table 15 reports the components under each alternative.

The pass-through of grain into food. The main results count every centner not delivered as food. If a farm had put part of it into its seed and fodder funds instead, the cost in people is smaller: at a pass-through of 0.6 the cost of the wedge and of the whole policy fall by about a third, and at a pass-through equal to the share of the residual that the farm's own division rule leaves for food when seed is short, 0.43, by about half (Table 15). The uniform claim and the misallocation of the wedge fall by less, by a tenth to a quarter, because the reallocations they measure move grain between farms rather than adding to the food of all. The timing of the relief falls roughly in proportion to the pass-through. The ordering of the components does not change under any pass-through.

The informed allocation. The benchmark allocates a fixed total so that every farm the total allows is held at the subsistence line. Allocating in proportion to each farm's harvest, with no regard to subsistence, lowers the cost of the uniform claim by two fifths, the cost of the misallocation of the wedge by half and the cost of the whole policy by seven per cent, because a proportional allocation does not hold the poorest farms at the line and the benchmark is itself less favourable (Table 15). The concentrated allocation, which pushes each district's shortfall onto as few farms as possible, leaves far fewer farm-years below the line, 16 at the statutory total and 24 at the office's, but preserves fewer people than the subsistence-first allocation, because the population response is concave and a farm pushed far below the line loses more members than several farms held near it. The count of farms below the line and the count of people therefore rank the allocations differently, and people are the measure reported.

By year. The 1934 figures apply the response estimated on the 1933 harvest to a year the panel does not observe. About two thirds of each component belongs to the 1933 harvest, and the whole of the cost of the relief's timing, since the 1934 harvest produced little relief (Table 15). Restricted to the 1933 harvest, the ordering of the components is unchanged.

8 Conclusion

In January 1933 the Soviet state replaced its claim on the assessed grain harvest of a collective farm with a fixed rent per sown hectare, the norm: one rate for every farm of a district, set before the sowing. A fixed rent cannot be moved by a report of the harvest, and it cannot follow the harvest;

Table 15: Robustness of the components.

Component	Main result	Pass-through 0.6	Pass-through 0.43	Proportional allocation	Concentrated allocation	1933 only
the policy as run, (b) against the benchmark	4,392	2,954	2,292	4,096	—	2,911
the uniform claim	707 [417; 1,019]	634	572	411 [238; 602]	502 [281; 754]	471
the wedge	3,685 [2,224; 5,179]	2,319 [1,403; 3,249]	1,719 [1,043; 2,404]	3,685	3,685	2,440
of which its misallocation	930 [547; 1,340]	772	704	422 [236; 636]	558 [289; 890]	618
the timing of the relief	964	640	479	964	964	964
farm-years below the line under (a) / (b)	92 / 161	110 / 161	121 / 162	83 / 144 under (g) / (d)	16 / 24 under (f') / (e')	69 / 124

Notes. People, from the response of Section 4.3; intervals ± 1.645 standard errors where reported by the run. The proportional and concentrated columns replace the subsistence-first allocation in the benchmark and in the informed allocation of the wedge; the wedge itself, (a) against (b), does not depend on the allocation rule. The 1933 column uses the 213 farm-years of the 1933 harvest, 64,052 members.

it taxes uniformly farms whose harvests differ. I measure what uniform taxation cost in people and divide the cost between the information the state lacked and the incentives of its district offices.

The measurement rests on the sequence of the agricultural year, which the statutes and orders of 1933 date step by step: the norm in the winter, applied to the area in the register; the sowing in May and its registration; the assessment of the standing crop in July and August, with the certification of the farms whose crop had failed; the collection from the first threshing to the autumn; the inspection of the seed in every granary against the register in the late winter; and the seed loan before the next sowing. Because the operations came in this order, the information available to each party at the time it acted follows from the sequence, and it can be written as a partition of the farms into the groups that party could distinguish. The centre, setting the norm before the sowing, distinguished farms by their registered area and by whether a station served them. The district office, collecting in the autumn, distinguished only the farms the commission had certified as failed, since no party above the farm measured a farm's harvest. The centre, lending in the spring, distinguished farms by their seed shortfall. No party observed the food of the members, and every correction came after the obligation it corrected had been collected. The accounts are consistent with these partitions. Within a district the delivery did not vary with the farm's own harvest and took two values, the statute on certified farms and one multiple on the rest; the multiple rose with the district's harvest and, between the two years, with its change. The seed loan followed the farm's own shortfall and not its neighbours'. The delivery is consistent with less misreporting than under the station's fee

and the flax contract, which remained claims on the harvest and show the signs the reporting game predicts; the accounts do not show that misreporting was absent.

The measurement rests, second, on the rules that the accounts allow to be estimated: the office's schedule; the farm's division of its residual grain between seed and food, a constant share when both are short; the relief, which filled the shortfall of every farm that was short; and the response of the number of members a year later to food per member. The model takes the norm and the district totals as given. The office's schedule can be rationalised as a corner solution, whatever weight the office placed on the district's future, and the relief as the restocking rule of a centre for which a restored hectare attracts the next year's rent. Together the rules convert any schedule of deliveries into food per member and members a year later. Each counterfactual replaces the observed deliveries by another schedule, farm by farm, with the harvest, the station's fee, the division rule and the response held fixed, and with the district's total held at the statutory total or at the total the office collected.

The policy as run cost 4,392 people over the two harvests, one member in twenty-eight, relative to the statutory total allocated on the farms' harvests with every farm held at subsistence where the total allowed. Each part of the cost is the difference between two schedules and is assigned to the constraint that separates them. The statutory total taxed uniformly per hectare against the same total allocated on harvests cost 707 people: the information the centre gave up when it wrote a fixed rent on the base it could verify. The office's total taxed uniformly per hectare against the same total on harvests cost 930 people: the office's information constraint, one multiple for every farm it could not tell apart. The office's total against the statutory total, both allocated on harvests, cost 2,755 people: the level of the wedge, the offices' collection above the norm, attributed to the incentive constraint at the district. Counting the seed lent to each farm before the 1934 sowing as food on the same farm in the autumn of 1933, with the total reduced by the loan, would have kept 964 people: the cost of learning from the seed stock in the late winter what the collection had done in the autumn.

The cost is that of the policy as run, given the norm. The grain that a crop share would have extracted is not identified, so the fixed rent is not compared with the contract system it replaced; the level of the district totals is not explained; the 1934 figures extrapolate the population response; and the response counts departures as well as deaths. A fixed rent on a base the state can verify removes the reporting game and gives up the information in the harvest: it taxes uniformly farms whose harvests differ, and any correction comes after the harvest, through the quantities the state can verify and at the time it verifies them. In subsistence agriculture the cost of uniform taxation falls where the rent crosses the subsistence line, because the population response to food is concave; the same norm cost half as much for the 1934 harvest, three fifths larger than that of 1933.

References

- Daron Acemoglu and Alexander Wolitzky. The economics of labor coercion. *Econometrica*, 79(2):555–600, 2011.
- P. Aghion and J. Tirole. Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29, 1997.
- D. W. Allen and D. Lueck. *The Nature of the Farm: Contracts, Risk, and Organization in Agriculture*. MIT Press, Cambridge, MA, 2002.
- R. Alonso and N. Matouschek. Optimal delegation. *Review of Economic Studies*, 75(1):259–293, 2008.
- M. Amador, I. Werning, and G.-M. Angeletos. Commitment vs. flexibility. *Econometrica*, 74(2):365–396, 2006.
- Kenneth J. Arrow, Theodore Harris, and Jacob Marschak. Optimal inventory policy. *Econometrica*, 19(3):250–272, 1951.
- D. P. Baron and R. B. Myerson. Regulating a monopolist with unknown costs. *Econometrica*, 50(4):911–930, 1982.
- Michael Carlos Best, Anne Brockmeyer, Henrik Jacobsen Kleven, Johannes Spinnewijn, and Mazhar Waseem. Production versus revenue efficiency with limited tax capacity: Theory and evidence from Pakistan. *Journal of Political Economy*, 123(6):1311–1355, 2015.
- Kim C. Border and Joel Sobel. Samurai accountant: a theory of auditing and plunder. *Review of Economic Studies*, 54(4):525–540, 1987.
- M. M. Coşgel and T. J. Miceli. Risk, transaction costs, and tax assignment: government finance in the Ottoman Empire. *Journal of Economic History*, 65(3):806–821, 2005.
- R. W. Davies and Stephen G. Wheatcroft. *The Years of Hunger: Soviet Agriculture, 1931–1933*. Palgrave Macmillan, Basingstoke, 2004.
- M. Ellman. Stalin and the Soviet famine of 1932–33 revisited. *Europe-Asia Studies*, 59(4):663–693, 2007.
- M. Halac and P. Yared. Commitment versus flexibility with costly verification. *Journal of Political Economy*, 128(12):4523–4573, 2020.

- L. Hurwicz. On informationally decentralized systems. In C. B. McGuire and R. Radner, editors, *Decision and Organization*, pages 297–336. North-Holland, Amsterdam, 1972.
- V. A. Il'inykh. Stanovlenie i funktsionirovanie kontraktatsionnoi sistemy v Sibiri (konets 1920-kh – nachalo 1930-kh gg.). In *Ezhegodnik po agrarnoi istorii Vostochnoi Evropy. 2014 god: Fiskal'naia politika i nalogovo-povinnostnye praktiki v agrarnoi istorii Rossii X–XXI vv.*, pages 357–368. Samarskii universitet, Samara, 2015. Reconstructs the contractation system from the archives of the apportionment; states that the volume a farm surrendered followed the procurement task and not the harvest, and that regional and district authorities could hand down more than they had received.
- V. A. Il'inykh and D. G. Simonov, editors. *Khlebnyi front Sibiri v 1930-e gg.: raionnye svodki. Sbornik dokumentov*. Sibprint, Novosibirsk, 2021. 489 pp. The preface, pp. 3–17, sets out the channels of centralized grain procurement, their legal history, and the carry-forward of unmet delivery and unmet station fees as arrears.
- V. V. Kondrashin. Khlebozagotovki v SSSR v gody pervoi piatiletki. In *Ezhegodnik po agrarnoi istorii Vostochnoi Evropy. 2014 god: Fiskal'naia politika i nalogovo-povinnostnye praktiki v agrarnoi istorii Rossii X–XXI vv.*, pages 369–382. Samarskii universitet, Samara, 2015. The regional criterion for the 1930 tasks was the growth of sown area, and the tasks were issued against areas planned to be sown.
- J.-J. Laffont and J. Tirole. *A Theory of Incentives in Procurement and Regulation*. MIT Press, Cambridge, MA, 1993.
- D. Landa and S. A. Tyson. Coercive leadership. *American Journal of Political Science*, 61(3): 559–574, 2017.
- Andrei Markevich, Natalya Naumenko, and Nancy Qian. The political-economy causes of the Soviet great famine, 1932–33. *Review of Economic Studies*, 92(5):3276–3305, 2025. Replication data: Zenodo 10983057.
- A. L. Men'shikova. Kolkhozy Urala v gody zaversheniia kollektivizatsii sel'skogo khoziaistva: istochnikovedcheskii aspekt, 2001. Ekaterinburg. A source study of the 1933–34 dynamic survey of the economic and financial condition of collective farms: its field dates, its instructions, and the survival of its primary forms for Sverdlovsk oblast in the regional archive (GASO f. 1812, op. 2).
- Iu. A. Moshkov. [title to be completed], 1966. Stub entry: publication details to be supplied by the author.

- R. B. Myerson. Incentive compatibility and the bargaining problem. *Econometrica*, 47(1):61–73, 1979.
- Natalya Naumenko. The political economy of famine: The Ukrainian famine of 1933. *Journal of Economic History*, 81(1):156–197, 2021.
- Robert M. Townsend. Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory*, 21(2):265–293, 1979.
- I. E. Zelenin. Osnovnye pokazateli sel'skokhoziaistvennogo proizvodstva v 1928–1935 gg. In V. K. Iatsunskii, editor, *Ezhegodnik po agrarnoi istorii Vostochnoi Evropy za 1965 g.*, pages 464–473. Nauka, Moscow, 1970. Table 1, “Grain yield on collective farms, centners per hectare”. Cited for the comparison of the dynamic surveys (TsGANKh f. 1562, *op.* 78) with the Central State Yield Commission, for the commission's instruction of 22 May 1933, and for the statement that the payment in kind to the machine-tractor stations came to be computed on the commission's figures.

A Proofs

Proof of Proposition 1. (i) Without the future term, the objective in (2) changes at the rate $b + q > 0$ per centner for $X_D < T_D$ and at the rate $b - q_O < 0$ for $X_D > T_D$, so it is maximised at $X_D = T_D$. The future term is non-increasing in every x_i , so it cannot make a total above T_D optimal, and by the condition on q it cannot make a total below T_D optimal. Hence $X_D = T_D$.

(ii) and (iii). Impose $X_D = T_D$ with multiplier λ_D and write the Lagrangian $\mathcal{L} = bT_D - \delta_O \sum_i [v(\bar{a}_i) - v(a'_i)] + \lambda_D(T_D - \mu_D A_V - \zeta_D A_F)$. Let χ_i equal one if farm i is short of seed and zero otherwise; then $\partial a'_i / \partial x_i = -(\varphi/\varsigma)\chi_i$. Write $\bar{\mathbb{E}}_V[z] = \sum_{i \in V_D} z_i n_i \bar{a}_i / A_V$ and $\bar{\mathbb{E}}_F[z]$ likewise, the averages per statutory centner. Then

$$\frac{\partial \mathcal{L}}{\partial \mu_D} = -A_V [\delta_O (\varphi/\varsigma) \bar{\mathbb{E}}_V[v'(a'_i)\chi_i] + \lambda_D], \quad \frac{\partial \mathcal{L}}{\partial \zeta_D} = -A_F [\delta_O (\varphi/\varsigma) \bar{\mathbb{E}}_F[v'(a'_i)\chi_i] + \lambda_D].$$

At an interior μ_D the first derivative is zero, so $\lambda_D = -\delta_O (\varphi/\varsigma) \bar{\mathbb{E}}_V[v'\chi]$. Substituting into the second,

$$\frac{\partial \mathcal{L}}{\partial \zeta_D} = -A_F \delta_O (\varphi/\varsigma) (\bar{\mathbb{E}}_F[v'(a'_i)\chi_i] - \bar{\mathbb{E}}_V[v'(a'_i)\chi_i]).$$

On the failed farms $\chi_i = 1$ and a'_i is below the area of every short farm in V_D ; since v is concave, $v'(a'_i)$ on a failed farm exceeds v' on every short farm in V_D , and $\chi_i \leq 1$ lowers the second average further. The bracket is therefore positive and $\partial \mathcal{L} / \partial \zeta_D$ is negative for every $\delta_O > 0$, so ζ_D falls to

its lower bound of one. The sign of $\partial \mathcal{L} / \partial \zeta_D$ does not depend on δ_O , which is (iii). With $\zeta_D = 1$ and $X_D = T_D$, $\mu_D = (T_D - A_F) / A_V$. $\square$

Proof of Proposition 2. (i) Let v_1 and v_2 be bounded. For any shortfall the two maximisations in (8) run over the same interval, and their objectives differ by at most $\delta \sup |v_1 - v_2|$; hence the operator on the right of (8) is a contraction of modulus δ in the supremum norm, and by Banach's theorem it has a unique bounded fixed point. If v is non-decreasing, then w in (9) is non-increasing in σ ; since $x(y, a)$ is non-decreasing in a and the shortfall $\sigma(y, a)$ is non-increasing in a (the residual rises with a because $x_a \leq y$), the operator maps non-decreasing functions into non-decreasing functions, and the fixed point is non-decreasing.

(ii) In (9) the maximand $\delta v(a') - (1 + \lambda)\zeta a'$ is single-peaked at a^* . Its maximiser over $[\bar{a} - \sigma/\zeta, \bar{a}]$ is $\max\{a^*, \bar{a} - \sigma/\zeta\}$. Substituting into $r = \sigma - \zeta(\bar{a} - a')$ gives $r^* = \max\{0, \sigma - \zeta(\bar{a} - a^*)\}$, which is (10).

(iii) The tolerance is zero if and only if $a^* = \bar{a}$, that is, if and only if the maximand is non-decreasing at $\bar{a}$, which under single-peakedness is $\delta v'(\bar{a}) \geq (1 + \lambda)\zeta$. Under full filling the farm returns to $\bar{a}$ every spring, so $v(a) = \mathbb{E}[x(y, a)] - (1 + \lambda)\mathbb{E}[\sigma(y, a)] + \delta v(\bar{a})$. The shortfall is continuous in the residual at both boundaries of the seed-constrained regime, so v may be differentiated under the expectation. Inside the regime the shortfall falls with the residual at the rate ϕ , and the residual rises with the area at the rate $y - x_a$; outside it the shortfall is zero. Hence $v'(\bar{a}) = \mathbb{E}[x_a] + (1 + \lambda)\phi \mathbb{E}[(y - x_a)\chi]$. $\square$

B The Reporting Game

Assumption 1 of Section 5 states that the obligatory delivery was computed from the registered sown area and that neither the farm's report of its harvest nor an assessment of it entered the claim. This appendix derives that property from the form of the claim and sets out the evidence in the accounts that is consistent with it. The quantity of grain the state required a farm to deliver, its claim on the farm, had to be computed from something the state could observe. Soviet procurement used two quantities for this purpose: the farm's harvest and the area the farm had sown. Nobody above the farm measured the harvest. It was either reported by the farm from its own barn or estimated by a commission from the standing crop. A claim computed from the harvest could therefore be influenced from both sides: the farm could understate its harvest, and the official who assessed it could overstate it. I call this interaction the reporting game. The sown area, by contrast, was recorded before the harvest and could be checked by inspection, so a claim computed from it could not be influenced through the harvest by either side. Section B.1 formalises this distinction in a model in which the claim is a weighted combination of the assessed harvest and the sown area;

when the weight on the harvest is zero, neither the farm nor the assessor gains from misreporting (Proposition B.1). Section B.2 describes the three claims that stood on a collective farm in 1933, one computed from sown area and two from the harvest, and Table B.1 summarises the evidence on each. Within a district, the grain delivery did not vary with the farm's own harvest (Section B.3). The fee paid to the machine-tractor station varied with the farm's harvest approximately in proportion, and the harvest from which it was computed exceeded the harvest recorded in the farm's barn (Section B.4). Under the flax contract, which remained a claim on output, the crop recorded in 1933 was a small fraction of the crop recorded the following year (Section B.5). The two claims on grain were set independently of each other (Section B.6); this independence is Assumption 2 of Section 5.

B.1 The Model

Consider one farm in one year. The farm harvests H centners of grain. The harvest is known to the farm but not to the state. Before the harvest, the farm sows $\bar{a}$ hectares; the area is entered in a land register, which officials can check. The state requires the farm to deliver T centners of grain. The claim T can be computed in two ways. Under a crop share, the farm delivers a fraction τ of its harvest. Under a land tax, the farm delivers a fixed quantity n per hectare sown; I call n the norm. To treat both forms in one framework, let the claim be a weighted combination of the two, with weight θ on the crop share:

$$T = \theta \tau \hat{H} + (1 - \theta) n \bar{a}, \quad 0 \leq \theta \leq 1. \quad (\text{B.1})$$

The harvest that enters equation (B.1) is $\hat{H}$, the harvest as assessed for the purpose of the claim, rather than the true harvest H . The two differ when the farm conceals part of its harvest, so that the harvest it reports falls short of the true harvest by $h \geq 0$ centners, and when the official who assesses the farm's harvest, whom I call the assessor, adds $o \geq 0$ centners to the assessment in order to raise the claim. Hence

$$\hat{H} = H - h + o. \quad (\text{B.2})$$

Both actions are costly. Concealing grain requires labour and hiding places and exposes the farm to prosecution; the law of 7 August 1932 made the theft of collective property a capital offence. Let $c_F(h)$ denote the farm's cost of concealing h centners. Overstating the harvest exposes the assessor to sanction; the law of 19 January 1933 prohibited any delivery obligation above the norm (art. 17), and the order of 20 June 1933 made the imposition of a supplementary plan a criminal offence (art. 5). Let $\kappa_{OO}(o)$ denote the assessor's cost of overstating the harvest by o centners, where $\kappa_O > 0$ measures the severity of the sanction. Both cost functions are zero at zero, increasing

and strictly convex, and their marginal costs rise from zero without bound.

The farm's payoff is the harvest it retains net of the cost of concealment, $H - T - c_F(h)$. The assessor is judged by the quantity collected. Let $B' > 0$ denote the value to the assessor of one additional centner delivered; the assessor's payoff is $B'T - \kappa_O c_O(o)$. The farm chooses h and the assessor chooses o . The marginal payoffs are

$$\frac{\partial}{\partial h} [H - T - c_F(h)] = \theta\tau - c'_F(h), \quad \frac{\partial}{\partial o} [B'T - \kappa_O c_O(o)] = B'\theta\tau - \kappa_O c'_O(o). \quad (\text{B.3})$$

The marginal benefit of concealment to the farm is $\theta\tau$: removing one centner from the assessed harvest lowers the claim by $\theta\tau$ centners. The marginal benefit of overstatement to the assessor is $B'\theta\tau$: adding one centner to the assessed harvest raises the claim by $\theta\tau$ centners, each worth B' to the assessor. Both marginal benefits are proportional to $\theta\tau$. Neither depends on the norm, on the sown area, on the true harvest, or on the other agent's action.

In a numerical example, suppose that a farm sows 100 hectares and harvests 500 centners. Under a land tax with a norm of 2 centners per hectare ($\theta = 0$), the farm owes 200 centners, whether it reports 500 centners or 400 and whether the assessor records 500 or 600. Under a crop share of one fifth ($\theta = 1, \tau = 0.2$), the farm owes 100 centners if it reports its harvest truthfully and 80 if it reports 400, so concealing 100 centners saves it 20. If the assessor records 600 centners, the farm owes 120, so the 100 centners added bring in 20. Under the land tax, neither side gains anything by misreporting the harvest. Proposition B.1 states the general result.

Proposition B.1. *Let $\tau > 0$, and let c_F and c_O satisfy the assumptions above. (i) If $\theta = 0$, the unique equilibrium is $h^* = o^* = 0$. (ii) If $\theta > 0$, the unique equilibrium is $h^* = (c'_F)^{-1}(\theta\tau) > 0$ and $o^* = (c'_O)^{-1}(B'\theta\tau/\kappa_O) > 0$; both are strictly increasing in $\theta\tau$.*

Proof. If $\theta = 0$, the farm's payoff is $H - n\bar{a} - c_F(h)$, which is strictly decreasing in h , and the assessor's payoff is $B'n\bar{a} - \kappa_O c_O(o)$, which is strictly decreasing in o . Each is maximised at zero, whatever the other agent does. If $\theta > 0$, each payoff is strictly concave in the agent's own action and has a positive derivative at zero, $\theta\tau$ for the farm and $B'\theta\tau$ for the assessor. Each first-order condition in (B.3) therefore has a unique root, and the root increases with $\theta\tau$ because marginal cost is increasing. In both cases the optimal action of each agent does not depend on the action of the other, so the equilibrium is in dominant strategies. $\square$

When the claim is computed entirely from sown area, neither the farm nor the assessor gains from misreporting the harvest, whatever the share rate and whatever the penalties. The farm's report does not affect what it owes, and the assessment is not a decision. When any part of the claim is computed from the harvest, both misreport, and the quantity collected is the joint result of

concealment by the farm and overstatement by the assessor. The higher the share rate, the larger both distortions.

The proposition concerns the harvest, not the land. A claim computed from sown area gives the farm an incentive to understate its area and an official an incentive to overstate it. The Soviet state inspected the register: the yield commissions verified the sown area (Section 2), and the sowing plan for the following year was based on the area last recorded. Across the 31 agricultural districts of Sverdlovsk oblast, the logarithm of a district's 1934 collective sowing plan rose with the logarithm of its recorded 1932 sown area with a coefficient of 0.86 (R^2 0.90). In the paper the registered area is treated as verified. The model covers a single year, whereas in the Soviet case an intertemporal consideration made a claim on the harvest still less attractive to the farm. Under the contract system in force until 1932, the quantity a farm had to deliver was apportioned to it on the basis of its expected harvest and could be raised during the year where the crop looked good (Section 2). A farm that revealed a good harvest could therefore expect a higher claim in the same year and in the next. Under the norm introduced in 1933, the following year's obligation was again computed from sown area, so a harvest revealed in one year did not raise the claim in the next.

The statutes of 1933 fixed a value of θ for each claim on a collective farm. The law of 19 January 1933 computed the grain delivery from the hectares of planned spring sowing and actual winter sowing at a norm per hectare (arts. 1 and 2): $\theta = 0$. The decree of 5 February 1933 computed the fee to the machine-tractor station as a percentage of the farm's gross harvest, with the gross harvest fixed for each farm by a district commission (decree no. 346, arts. 9 to 12): $\theta = 1$, with $\hat{H}$ the commission's figure and τ the percentage due for the operations performed, about 9 per cent of the gross harvest on the farms in the panel (Section 3). The decree of 11 August 1933 kept flax under contract: the farm owed the procurement agency Zagotlen a quantity of fibre and seed fixed in a contract enforceable in court, and kept whatever remained after delivery (decree no. 370, arts. 1 to 3). Flax was not brought under obligatory delivery and no norm per hectare was set for it, so the claim on flax was computed from the crop: $\theta = 1$. Proposition B.1 therefore predicts that neither the farm nor the assessor misreported under the grain delivery, and that both did under the station's fee and under the flax contract, more under the flax contract, which took a far larger share of the crop, than under the fee (Section B.5). Since all three claims were levied on the same farms in the same year, the prediction can be compared with the accounts within a farm-year, holding the harvest, the weather, the district and the officials constant. Table B.1 summarises the evidence.

The form of the 1933 law corresponds to the optimal contract in models of costly state verification (Townsend, 1979; Border and Sobel, 1987), in which the principal sets a fixed payment and verifies the agent's circumstances only when the agent reports that it cannot pay. In 1933 the fixed payment was the norm, and verification took the form of the yield commissions' certification of a failed harvest (Section 2); the treatment of the farms so certified is examined in Sections 5.4 and

6.2.

Sections B.2 to B.6 to be written.

Table B.1: Three claims on one farm: the evidence on each.

to be built with Section B.2

C Construction of the Counterfactuals

To be written.

D Data Appendix

To be written.